5DATA004C.2

Data Science Project Lifecycle

GROUP COURSEWORK

Project: KJ Marketing Customer Segmentation

Project's GitHub Repository

Team: Insight Syndicate

Group A2

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IIT NO.	20231818	20232052	20222221	20232555

Table of Contents

Acknowledgements

As Team Insight Syndicate, we would like to express our sincere gratitude to our lecturers and project supervisors, Mr Fouzul Hassan and Ms Sapna Dissanayaka, for their mentorship and guidance throughout this module. Their valuable insight and feedback were instrumental in successfully creating a well-made analytical solution and covering the project scope.

We are also grateful to the team at Octave for collaborating with this module, allowing us to work on an actual data set and problem given by industry experts. This exposure is truly beneficial for us as it will allow us to gain clear insight into the practices in the industry.

In addition, we extend our appreciation to our peers and colleagues who worked with us, provided peer feedback, and were supportive throughout this project. It has allowed us to provide a high-quality output. Our heartfelt thanks go to our friends and family for their constant support and encouragement.

Introduction

This project relates to KJ Marketing, a project of Octave that has highlighted the noticeable ineffectiveness of its marketing strategies in its current customer base. To resolve this issue, a four-member team was established to identify and analyse historical sales data to help KJ Marketing classify customers into their relevant customer segments. To begin, the firm's three product types (Dry items, fresh, and luxury items) and the regions in which they are sold will be analysed, spanning all 22 outlets available.

Background & Problem Domain

Company Overview:

KJ Marketing is a leading retail supermarket chain in Sri Lanka, operating 22 outlets across urban and suburban regions. The company offers diverse products, including dry goods, fresh produce, and luxury items, catering directly to end consumers. KJ Marketing is a project of **Octave**, the Data and Advanced Analytics Division of the John Keells Group, a diversified conglomerate with interests in multiple industry verticals.

Problem Statement:

Traditional marketing strategies have proven ineffective in engaging KJ Marketing's evolving customer base in recent years. The company aims to adopt a personalised marketing strategy tailored to individual customer preferences to enhance its marketing efforts. The goal is to develop an analytical solution that accurately classifies new customers into predefined customer segments based on their purchasing behaviour.

Project Scope:

The project analyses historical sales data provided by KJ Marketing, including average monthly sales per customer across different product categories (dry goods, fresh produce, and luxury items). The objective is to classify new customers into relevant segments and address key questions related to data preprocessing, feature selection, model training, and evaluation.

Key Tasks:

1. **Data Preprocessing:** Address missing values, duplicates, and outliers.
2. **Feature Selection:** Determine relevant features for classification.
3. **Feature Scaling/Normalization:** Apply techniques to improve model performance.
4. **Encoding Strategies:** Implement encoding methods for model input.
5. **Feature Correlation:** Analyze relationships between features and the target variable.
6. **Target Variable Interpretation:** Define and interpret customer segments.
7. **Algorithm Selection:** Choose and justify the final algorithm.
8. **Model Training Challenges:** Address issues like overfitting.
9. **Cluster Definition:** Define and name classified clusters.
10. **Marketing Strategy Enhancement:** Explain how the solution improves marketing effectiveness.

Customer Segments:

Six customer segments have been identified based on purchasing behaviour:

1. **Practical Bulk Buyers:** High sales in dry goods, moderate sales in luxury and fresh items.
2. **Fresh-Only Buyers:** High sales in fresh produce, low sales in dry and luxury items.
3. **Premium All-Round Buyers:** High and balanced sales across all categories.
4. **Necessity Bulk Buyers:** High sales in dry goods, low sales in fresh and luxury items.
5. **All-Round Buyers:** Balanced sales across all categories, but not as high as premium buyers.
6. **Selective Fresh Buyers:** High sales in fresh produce, moderate sales in luxury items, and low sales in dry goods.

Diagrams/Models:

To explore the problem domain, suitable diagrams and models such as box plots, scatter plots, and correlation matrices are used to visualise data relationships and patterns. These visualisations help understand the data's structure, quality, and key characteristics, aiding in identifying relevant features and developing the analytical solution.

Machine Learning Models:

Several machine learning models were considered for this project:

1. **Logistic Regression:** Estimates the probability of a customer belonging to a specific segment using a logistic function.

2. **K-Nearest Neighbors (KNN):** Classifies new customers by finding the K closest data points in the training dataset.

3. **Decision Tree:** Splits data based on feature values, forming a tree-like structure to make predictions.

4. **Random Forest:** An ensemble method combining multiple decision trees to increase accuracy and reduce overfitting.

5. **Gradient Boosting:** An ensemble method where sequential decision trees correct errors made by previous trees to improve accuracy.

This comprehensive approach aims to enhance KJ Marketing's ability to engage customers through personalised marketing strategies, ultimately improving customer satisfaction and business performance.

Expected deliverables:

1. Analytical Solution
    - A machine learning model capable of classifying new customers into six segments.
    - The model is tested on the test.csv dataset and outputs a CSV file with the predicted cluster category for each customer.
2. Final Project Report
    - Report including all the technical and business aspects of the project.
3. Technical Code
    - Complete source code for data preparation, modelling and prediction.
4. Presentation Slides
    - A clear and concise presentation to summarise and present the full project.

Project Planning

Team Roles

Iffath Saleem - Project Manager

Govindu Sathruwan - Data Scientist

Shahik Shiyam -Data Analyst

Abdullah Sheriffdeen - Business Analyst

Work Breakdown Structure

TASK	MAIN ASSIGNEE	SECONDARY ASSIGNEE
Draft Proposal		
Initial Dataset Exploration	Shahik	Abdullah
Project Objectives	Abdullah	Abdullah
Project Background	Govindu	Iffath

Challenges and Considerations	Govindu	Iffath
Project Plan - Gantt Chart	Iffath	Shahik
WBS	Iffath	Govindu
Data Cleaning		
Data Understanding and Exploration	Shahik	Abdullah
First EDA (Before Cleaning): Univariate Multivariate	Abdullah	Shahik
Handling Missing Values Handling duplicate values Data Validation Handling Outliers	Govindu	Iffath
Feature Engineering	All	All
Second EDA (After Cleaning): Univariate	Iffath	Shahik

Multivariate		
Model Creation		
Logistic Regression - Baseline Model	Iffath	Govindu
KNN - Baseline Model	Iffath	Govindu
Decision Tree - Baseline Model	Iffath	Govindu
Random Forest - Baseline Model	Iffath	Govindu
Gradient Boosting - Baseline Model	Iffath	Govindu
Data Scaling	Govindu	Shahik
Logistic Regression - Model	Iffath	Shahik
KNN - Model	Abdullah	Shahik
Decision Tree - Model	Shahik	Abdullah
Random Forest - Model	Govindu	Abdullah
Gradient Boosting - Model	Govindu	Abdullah

Final Model Selection	All	All
Presentation & Report		
Draft Report Compilation	All	All
Presentation Preparation	All	All
Final Report Compilation	All	All

KJ marketing project Gantt Chart

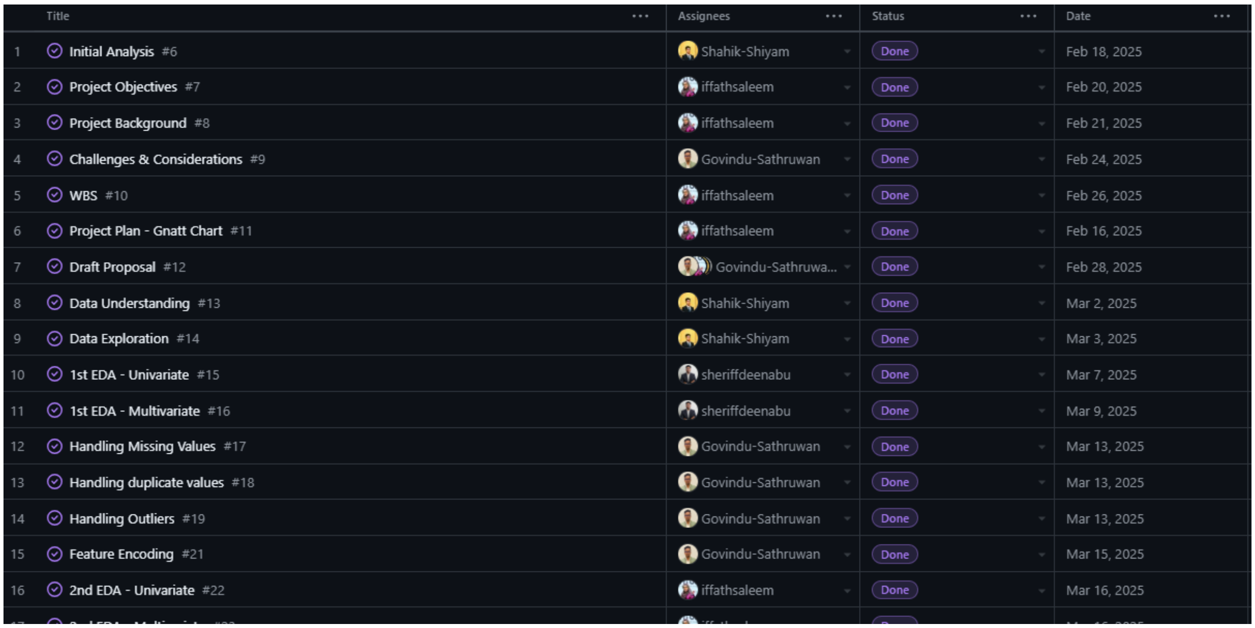
TIMELINE	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7
Initial Analysis	COMPLETED						
Project Objectives	COMPLETED						
Project Background	COMPLETED						
Challenges & Considerations	COMPLETED						

WBS	**COMPLETED**						
Project Plan - Gnatt Chart	**COMPLETED**						
Draft Proposal		**COMPLETED**					
Data Understanding		**COMPLETED**					
Data Exploration		**COMPLETED**					
1st EDA - Univariate		**COMPLETED**					
1st EDA - Multivariate		**COMPLETED**					
Target variable Analysis		**COMPLETED**					
Handling Data Type Issues			**COMPLETED**				
Handling Missing Values			**COMPLETED**				

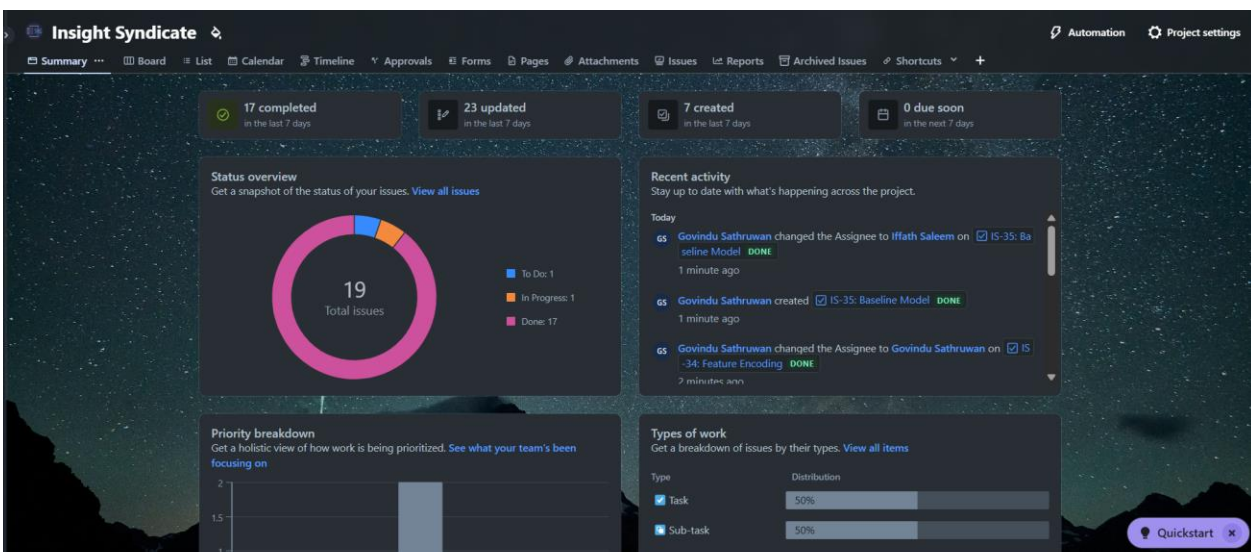
Handling duplicate values			**COMPLETED**				
Handling Outliers			**COMPLETED**				
Data Validation			**COMPLETED**				
Feature Engineering			**COMPLETED**				
Feature Encoding			**COMPLETED**				
2nd EDA - Univariate			**COMPLETED**				
2nd EDA - Multivariate			**COMPLETED**				
Baseline Model				**COMPLETED**	**COMPLETED**		
Feature Scaling				**COMPLETED**	**COMPLETED**		
Actual Model				**COMPLETED**	**COMPLETED**		
Hyperparameter					**COMPLETED**	**COMPLETED**	

Tuning							
Model Evaluation						COMPLETED	
Final Model selection						COMPLETED	
Business Recommendations & Strategies						COMPLETED	
Progress Report and Presentation Compilation						COMPLETED	
Exporting Final Model							COMPLETED
Running Model with test csv							COMPLETED
Final Report and Presentation Compilation							COMPLETED

- Jira has been selected as the project management tool for this project since this is the industry standard, and GitHub will be used for code collaboration. In addition we also used Github Project to assign tasks to team members.



	Title	Assignees	Status	Date
1	Initial Analysis #6	Shahik-Shiyam	Done	Feb 18, 2025
2	Project Objectives #7	iffathsaleem	Done	Feb 20, 2025
3	Project Background #8	iffathsaleem	Done	Feb 21, 2025
4	Challenges & Considerations #9	Govindu-Sathruwan	Done	Feb 24, 2025
5	WBS #10	iffathsaleem	Done	Feb 26, 2025
6	Project Plan - Gnatt Chart #11	iffathsaleem	Done	Feb 16, 2025
7	Draft Proposal #12	Govindu-Sathruwa...	Done	Feb 28, 2025
8	Data Understanding #13	Shahik-Shiyam	Done	Mar 2, 2025
9	Data Exploration #14	Shahik-Shiyam	Done	Mar 3, 2025
10	1st EDA - Univariate #15	sheriffdeenabu	Done	Mar 7, 2025
11	1st EDA - Multivariate #16	sheriffdeenabu	Done	Mar 9, 2025
12	Handling Missing Values #17	Govindu-Sathruwan	Done	Mar 13, 2025
13	Handling duplicate values #18	Govindu-Sathruwan	Done	Mar 13, 2025
14	Handling Outliers #19	Govindu-Sathruwan	Done	Mar 13, 2025
15	Feature Encoding #21	Govindu-Sathruwan	Done	Mar 15, 2025
16	2nd EDA - Univariate #22	iffathsaleem	Done	Mar 16, 2025



- Furthermore, as the coding platform, we have decided to use Google Colab for this project after connecting Collab to the project's GitHub Repository.

Potential Risks & Mitigation Plan

Every project or business plan faces the potential risks of difficulties that can alter and stop progress. Therefore, it is essential as a group to develop a mitigation plan to minimise or even eliminate these risks, allowing a smooth flow of progression throughout the timeline. Below are some identified risks and their mitigation plans.

As this is one of the most significant projects the team has experienced, the risk of team members not fully understanding the project overview and the technicalities within to solve the project is high. Therefore, to mitigate this and make sure all team members are on the same page, regular team meetings will be conducted along with clear documentation of project progress and goals; this will allow us to implement techniques like SMART (Specific, Measurable, objective, Time-bound). (funds for NGOs, 2024)

Moreover, as this project spans around 7 weeks, there are risks of certain members not making it to every group meeting due to unforeseen circumstances during the extended timeline. To ensure this doesn't happen and a bottleneck is prevented, backup responsibilities are assigned, and access to files is shared. This allows team members to cover for each other and ensure progress is continued temporarily. (Carnegie Mellon University, no date)

In addition to this, there are possibilities of device malfunctions and technical breakdowns. Cloud-based applications such as Google Collab, Google Docs, and GitHub can be used to ensure this does not hinder progress. It can also help mitigate data loss as it will be saved in the cloud. This can be a huge safety net. (Janosevic, 2023)

Furthermore, as this is a lengthy project with many technicalities that members may find too hard to cope with and absorb instantly, there is a risk of burning out and a sense of demotivation creep in. To mitigate this, we can categorise what needs to be done immediately and distribute the workload evenly. We can also break tasks down and set realistic deadlines. By setting up small goals to achieve, it will allow us to make significant progress in the long run. (Johansson, 2024)

Data Understanding and Preprocessing

Data Exploration

This process is the first crucial step before applying modelling techniques or making decisions, as it helps identify patterns, potential issues and areas for further investigation. Initial data exploration is the first task of analysing a dataset to understand its structure, quality and key characteristics.

Initial data exploration was conducted on both test and train data to ensure data quality, model performance and validity of results.

We begin by obtaining the summary of the dataset. This is important as it further provides information about the features within the dataset and how many features are available alongside what the data type of each feature is. For instance, we have six features for the train dataset and five features for the test dataset through the output provided

```
test.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40749 entries, 0 to 40748
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Customer_ID   40749 non-null  int64
 1   outlet_city   40749 non-null  object
 2   luxury_sales  40749 non-null  object
 3   fresh_sales   40749 non-null  object
 4   dry_sales     40749 non-null  object
dtypes: int64(1), object(4)
memory usage: 1.6+ MB
```

```
train.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 774155 entries, 0 to 774154
Data columns (total 6 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Customer_ID       774153 non-null  float64
 1   outlet_city       774153 non-null  object
 2   luxury_sales      774120 non-null  object
 3   fresh_sales       774114 non-null  object
 4   dry_sales         774125 non-null  object
 5   cluster_catgeory  774154 non-null  object
dtypes: float64(1), object(5)
memory usage: 35.4+ MB
```

From there, we proceed to find the descriptive statistics, which include information such as the count, mean, min and max, just to name a few. This provides valuable insight into figures that can be used to compare for data analysis and preprocessing tasks like data cleaning.

```
[4] test.describe()
```

	Customer_ID
count	40749.000000
mean	20375.000000
std	11763.367396
min	1.000000
25%	10188.000000
50%	20375.000000
75%	30562.000000
max	40749.000000

```
train.describe()
```

	Customer_ID
count	7.741530e+05
mean	1.038708e+07
std	2.234795e+05
min	1.000000e+07
25%	1.019354e+07
50%	1.038708e+07
75%	1.058062e+07
max	1.077415e+07

We then retrieve the dimensions of the dataset, which, in simple terms, identifies how many rows and columns are available in our dataset. This helps indicate how big the dataset is.

```
[148] test.shape
(40749, 5)
```

```
train.shape
(774155, 6)
```

The next step was to identify the unique columns in the dataset, which is crucial during EDA and data cleaning. This is because it helps identify key focuses, such as duplicate entries or missing values, which need to be attended to during the data cleaning process.

```
unique_values_test = test.apply(pd.Series.unique)
print(unique_values_test)
```

```
Customer_ID     [33574, 10089, 38329, 11376, 12410, 826, 10364...
outlet_city     [batticaloa, Batticaloa, Colombo, Dehiwala-Mou...
luxury_sales    [2686.5, 1717.56, 854.04, 1638.12, 1039.09, 12...
fresh_sales     [3582, 2576.34, 1242.24, 2320.67, 1518.67, 149...
dry_sales       [12537, 9446.58, 5201.88, 9282.68, 5435.24, 68...
dtype: object
```

```
[ ] missing_values = test.isnull().sum()
    print(missing_values)
```

```
Customer_ID     0
outlet_city     0
luxury_sales    0
fresh_sales     0
dry_sales       0
dtype: int64
```

```
missing_values_train = train.isnull().sum()
print(missing_values_train)
```

```
Customer_ID          2
outlet_city          2
luxury_sales        35
fresh_sales         41
dry_sales           30
cluster_catgeory     1
dtype: int64
```

Additionally, we also had to identify how many missing values existed in our test and train datasets, and as can be seen below, the test dataset had no missing values, whereas the train dataset had a total of one hundred and eleven missing values, which will again have to be resolved in the data cleaning process.

Similar to missing values, identifying how many duplicate entries are found in the dataset is an important step as it is again another key element of the data cleaning process and is essential in ensuring the dataset is tidy for EDA.

```
num_duplicates1 = test.duplicated().sum()
num_duplicates1
```

```
0
```

```
num_duplicates_train = train.duplicated().sum()
num_duplicates_train
```

```
0
```

Data Cleaning

We looked into several cleaning procedures with several implementation methods for the data cleaning process for multiple dataset issues.

Data Type Handling

In the train dataset, the first step in the data cleaning process was to convert the features to the correct data type. Therefore, the sales features were turned into float data types, while the other three features were kept as object data types since they were classes of categories. When the sales features were turned to numeric, the invalid data points that could not be converted were turned into missing values.

On the other hand, in the test dataset, the issue that was identified was that some features contained text values input instead of numerical values. To resolve this, we located the customer_ID with the problem and assigned the correct integer value to its place, thereby resolving it. This could be done manually since only four tuples of such an issue were found in the dataset.

Missing Value Handling

In the train dataset, for the missing values in the sales features, since there was skewness in the dataset, it was decided to implement a median imputation system to fill in the missing values for the sales features. For the other variables (Customer_ID and outlet_city) where median imputation seemed impractical, we found the percentage of missing in proportion to the entire dataset and since they were both random missing values and were less than 0.001% of the dataset, a decision to drop such missing values was made.

Misspellings and duplicate handling

From there, we proceeded to identify all the available outlet cities, which were then used to fix duplicates and spelling mistakes within the dataset. For instance, the city Batticaloa had two entries, one in opening simple letters and the other in opening capital letters. This was fixed by replacing all “Batticaloa” entries with “Batticaloa”. A similar approach was implemented for all duplicated and misspelt cities.

In the train dataset, no duplicate values were found. However, there were a few erroneous entries that were replaced with the correct value using the “str.replace” function. Similarly, under cluster_category, there should be only six categories, but the dataset has categories that are not in this range. Due to that, the “.drop” function was used to remove the error categories since all of the invalid categories had only one row of data, which helped clean the invalid data further.

Handling outliers

In the case of outliers, for both test and train datasets, “luxury_sales” was the only feature containing outliers, and the group mutually decided to leave it as it is since luxury goods and services have high prices and could generally be considered to be lot more expensive hence the plots above the above boundary.

Feature Engineering

Feature Encoding

Finally, the last step was to make use of label encoding. As one of the features, “outlet city”, was a categorical variable, we used label encoding to turn it into a numeric format so that the machine learning models could process it effectively. Since it is a non-binary categorical variable, this encoding was best suited for the dataset. Furthermore, since label encoding does not increase the dimensionality of the dataset, it would not negatively impact our models. This was also an essential task as it helps the machine learning algorithms process data in a more effective manner by converting the variable from text to numerical.

Feature Scaling

Next, data scaling was performed on the train data, specifically on the three sales features, to bring all three into similar magnitude for better modelling. A Robust scaling technique was employed to perform this scaling since the data was skewed and had outliers in luxury sales. The robust scaler excels at handling these conditions. A robust scaler uses the median and IQR to get the normalised values.

$$Xscaled=(X-Q2)/(Q3-Q1)$$

luxury_sales	fresh_sales	dry_sales	luxury_sales	fresh_sales	dry_sales
1209.6	756.0	5292.0	0.857034	0.041937	1.631090
1590.12	1060.08	6007.12	-0.003695	-0.156121	1.057081
2288.88	1481.04	9155.52	-0.770778	-0.418864	0.268678
2274.94	1739.66	9099.76	-0.074264	-0.206474	1.026639
2345.49	2069.55	9243.99	-0.606394	-0.364423	0.312022
...	...	...	...	...	...
3893.4	3893.4	3448.44	1.949650	0.156019	0.029930
6095.86	5557.99	6275.15	1.306009	-0.071865	-0.122086
5121.42	4820.16	4669.53	1.880468	0.114708	-0.010279
6311.76	6311.76	5940.48	2.302562	0.160308	0.103720
5833.5	6027.95	6611.3	2.212309	0.166043	0.084849

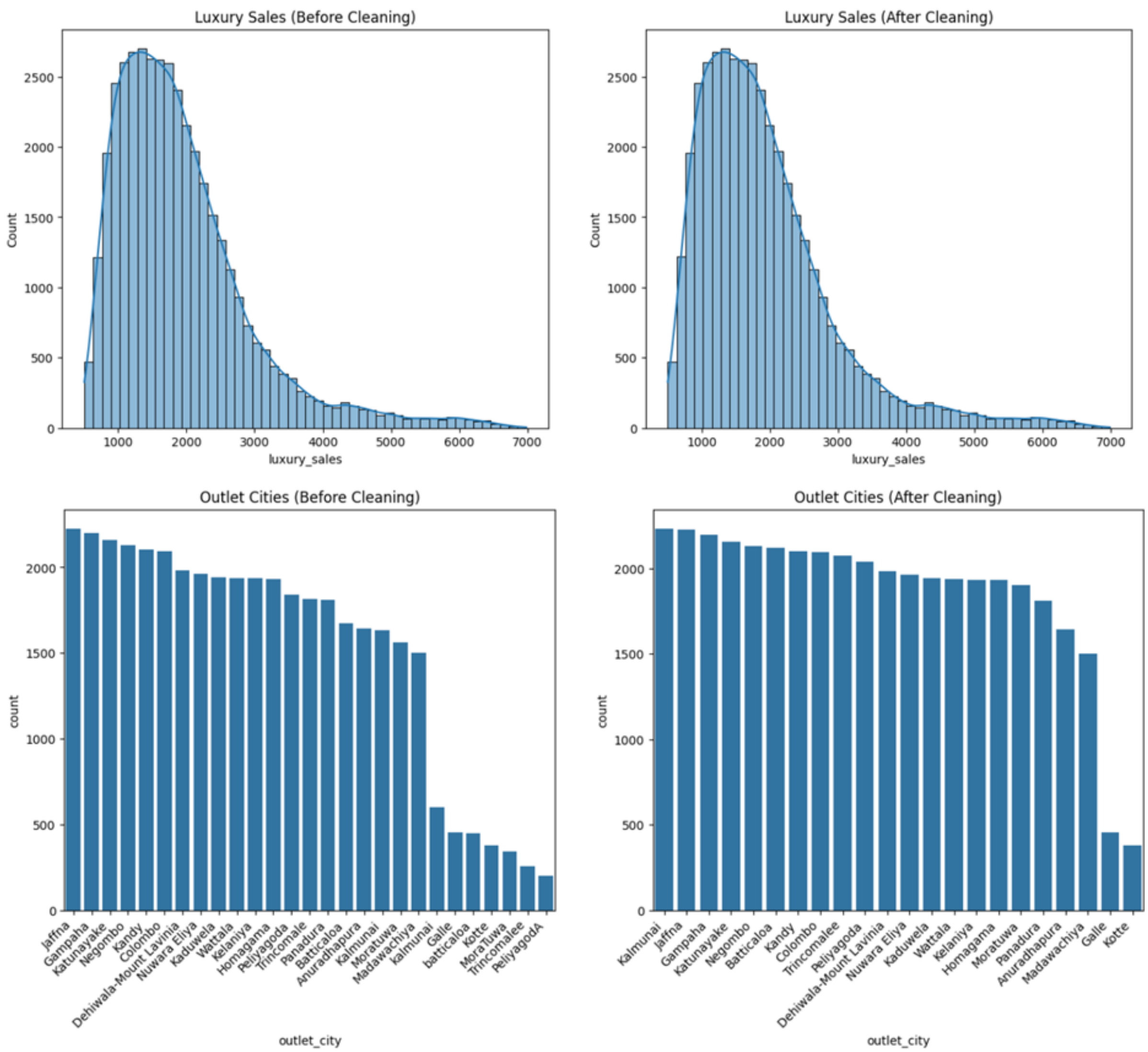
Feature Selection

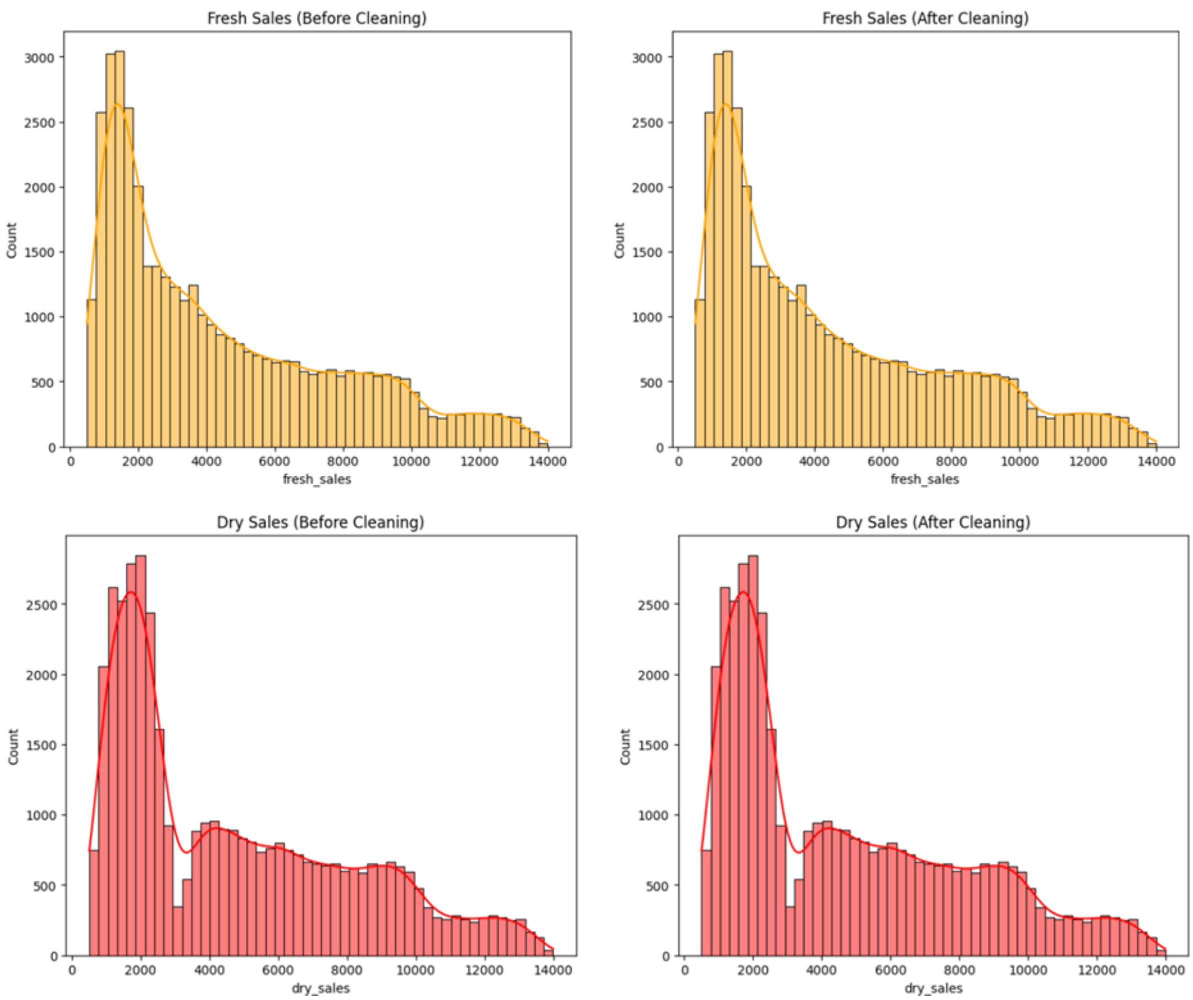
In the case of this problem, all features of the dataset were chosen to be worked on during the preprocessing and EDA steps. However, when it came to creating the machine learning models, only the sales features and the outlet city encoded features were used. Customer_ID was not used, as the specificity of the variable provided no predictability that could be considered for the model and would simply contribute to overfitting. Similarly, since machine learning models deal better with numeric data, the outlet city encoded feature generated using label encoding was used instead of the original outlet city feature.

The three sales features are the key to segmenting customers for this problem, as KJ Marketing requires performing targeted marketing based on customer spending patterns to obtain the best results from their marketing campaigns. Outlet cities also contribute to the segmentation of the customer base, but based on the EDA and the importance of features extracted from ML models like Random Forest and Gradient Boosting, sales features carry more weight in customer segmentation than outlet cities.

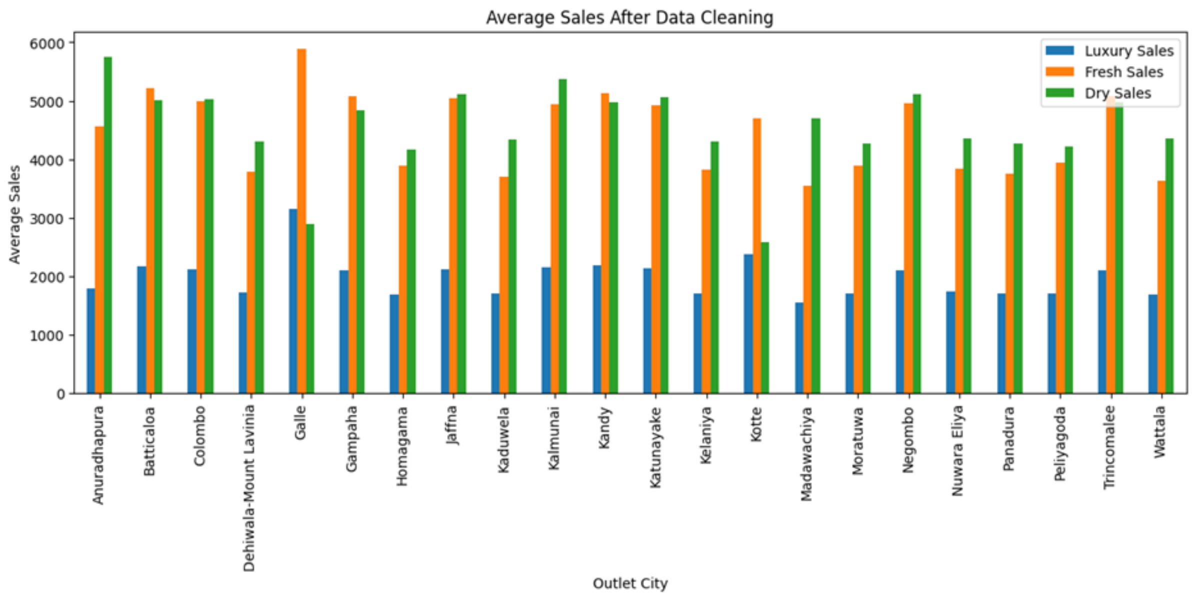
Exploratory Data Analysis (EDA)

EDA Before & After Data Cleaning - Test CSV

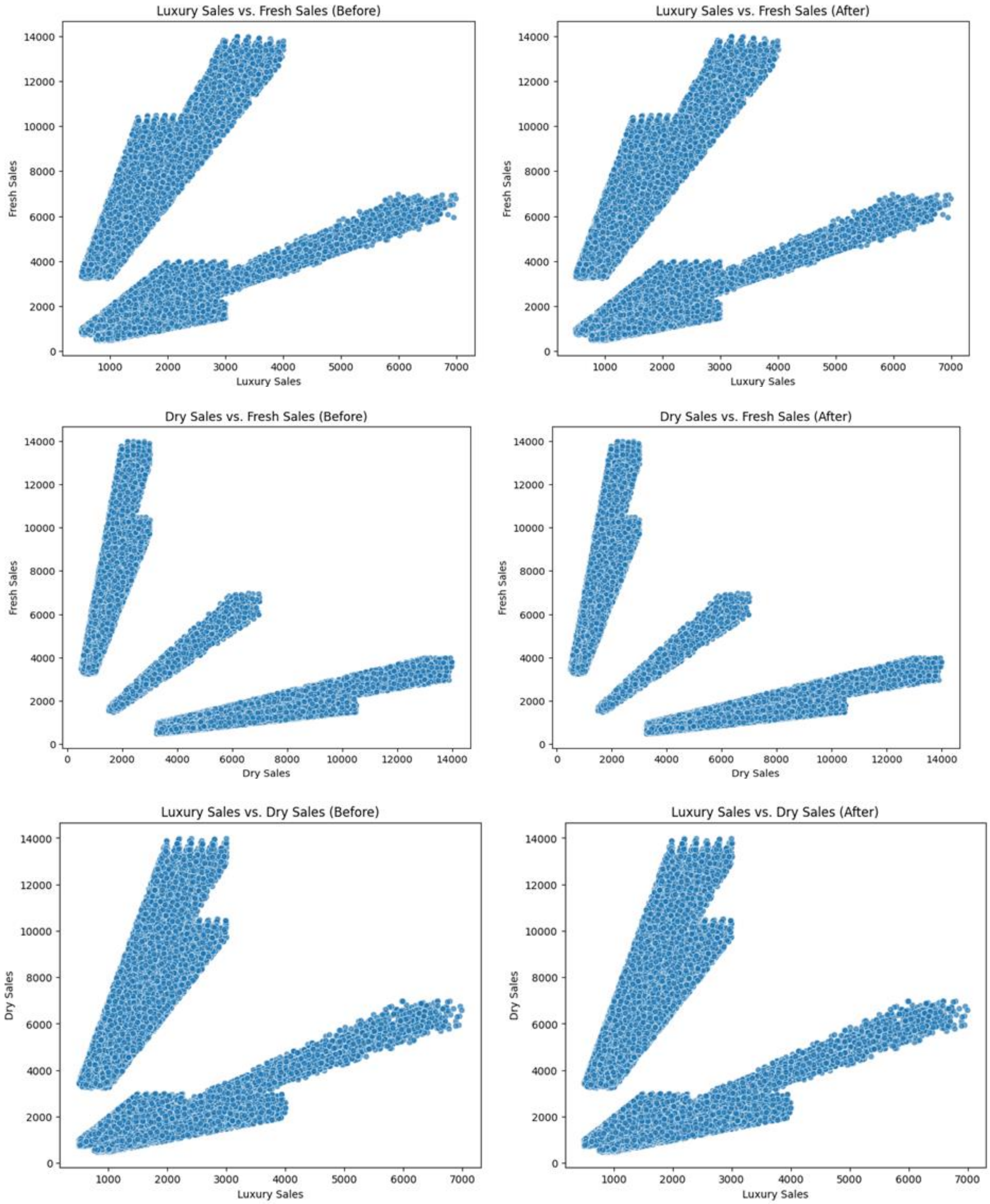




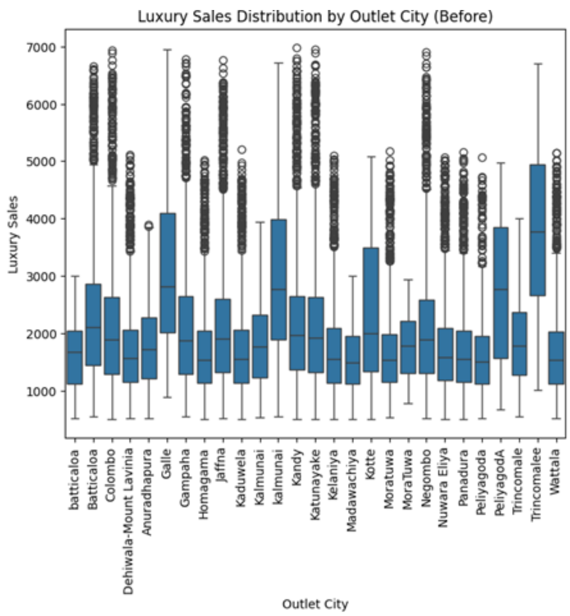
The above graphs compare the different columns in the dataset before and after data cleaning, showing that some data may have been removed while maintaining a similar distribution among the categories.

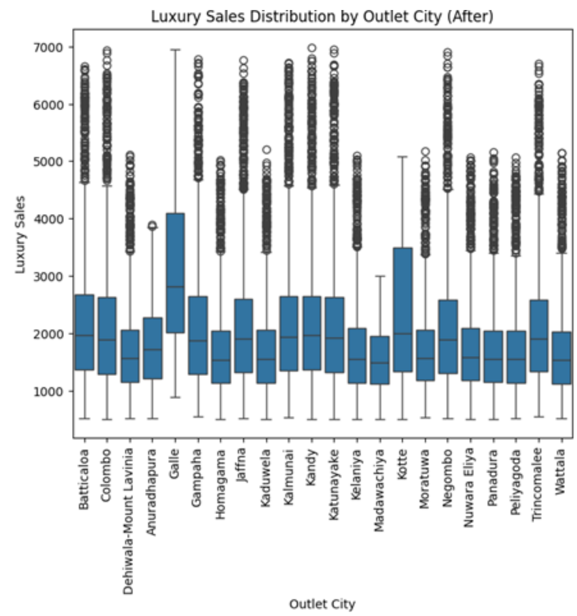


After cleaning the data, the average sales values demonstrated changes throughout different cities. It has new averages in other cities, and the locations of towns on the x-axis display the same sequence as in the "Outlet Cities" graphs.

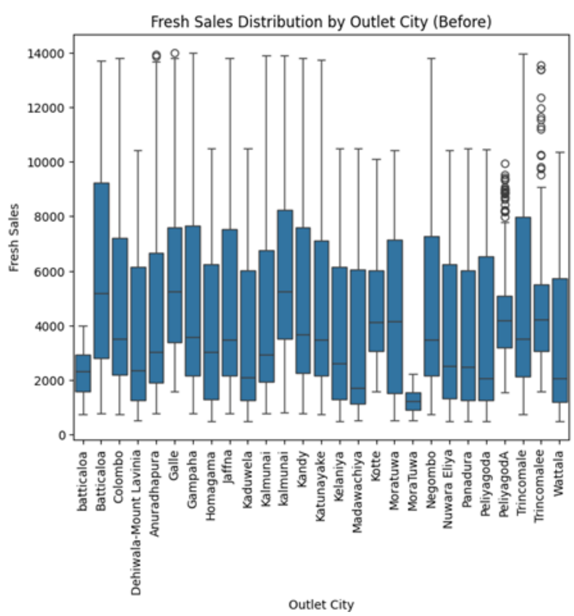


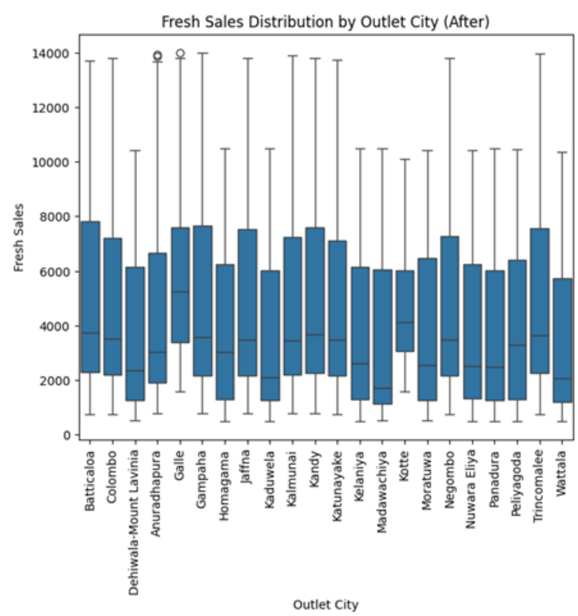
The scatter plots before and after cleaning might have slight differences but are not significant notable differences.

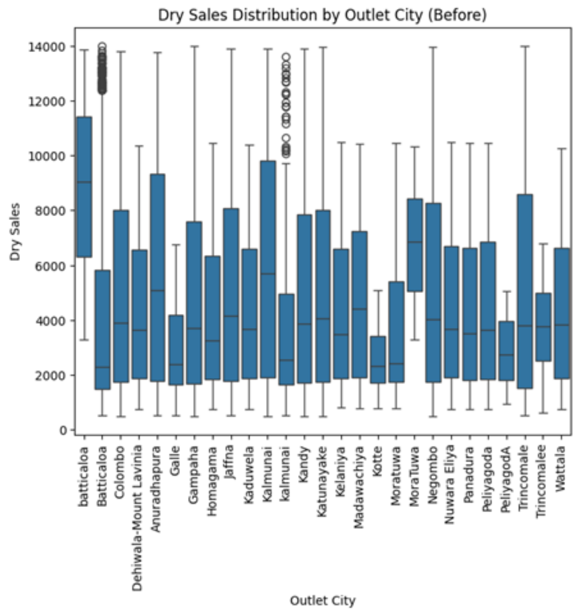


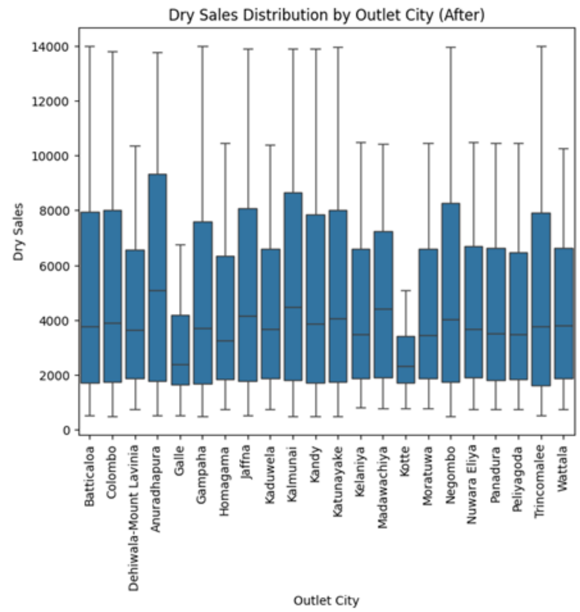


The data cleaning didn't impact the luxury sales category much; they are meant to have outliers due to the high prices.



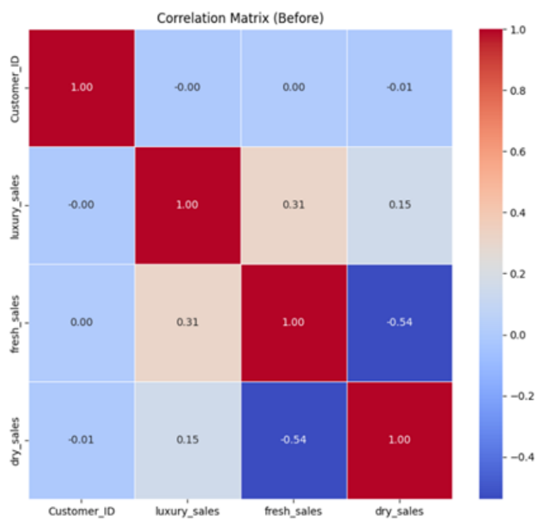


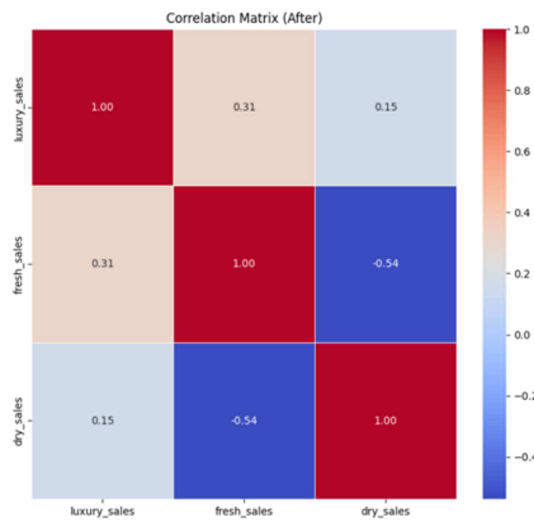




Inconsistencies in city names are shown before cleaning the distribution of fresh and dry sales across different cities.

Fresh and dry sales distribution by the city is refined, and city names are standardised after data cleaning.

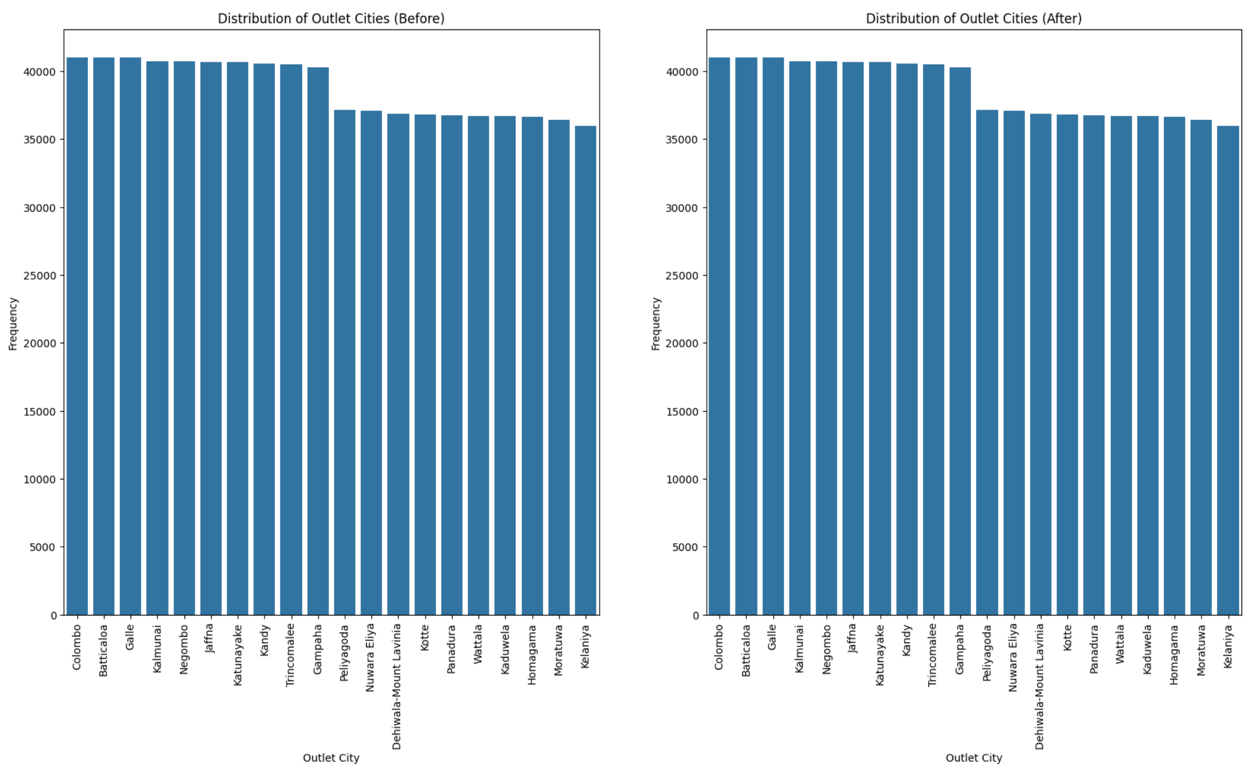




Before cleaning, The correlation matrix displays the correlation coefficients between variables (luxury sales, fresh sales, and dry sales).

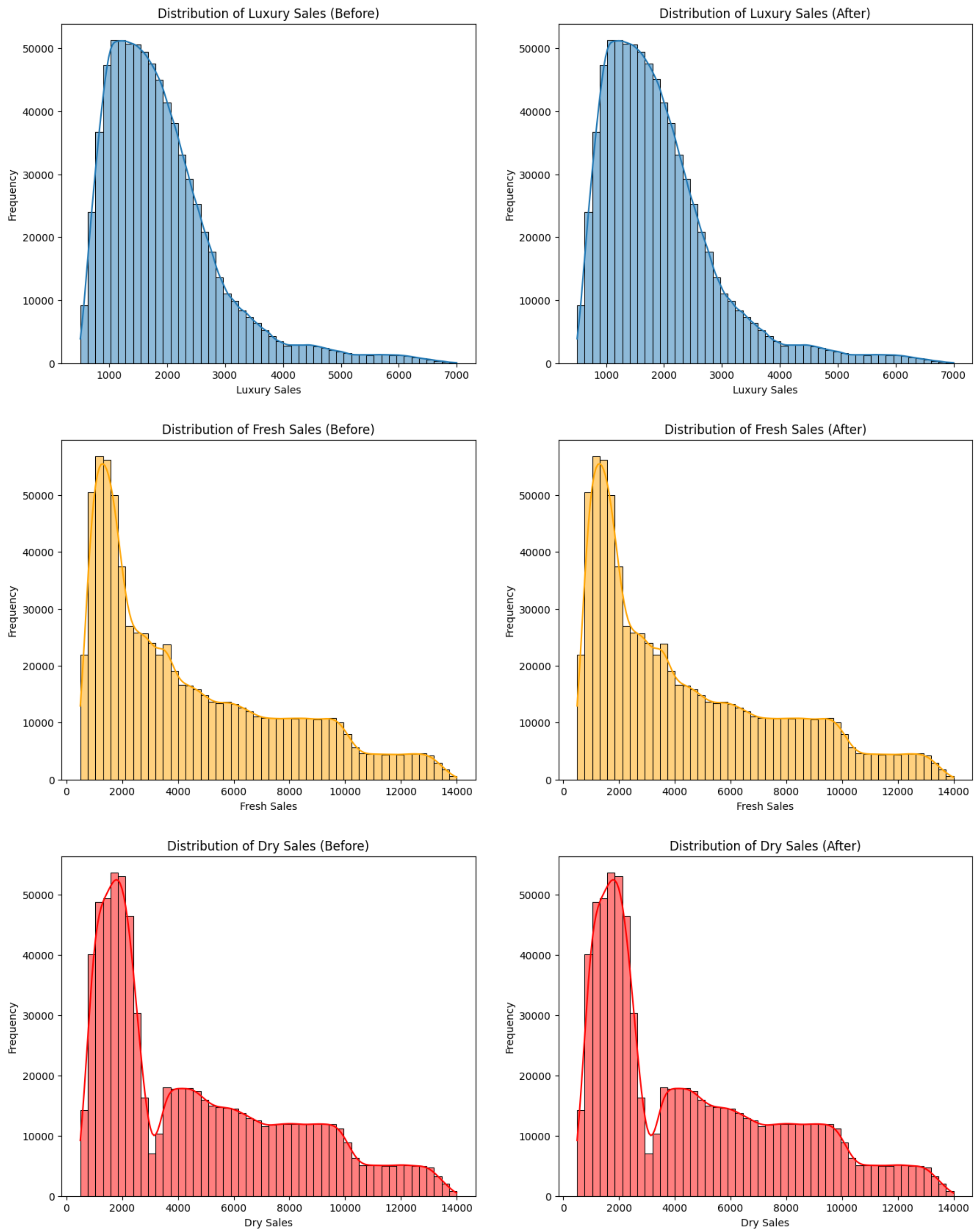
After data cleaning, the correlation coefficients between the same variables are altered, indicating that the relationships between the variables have changed.

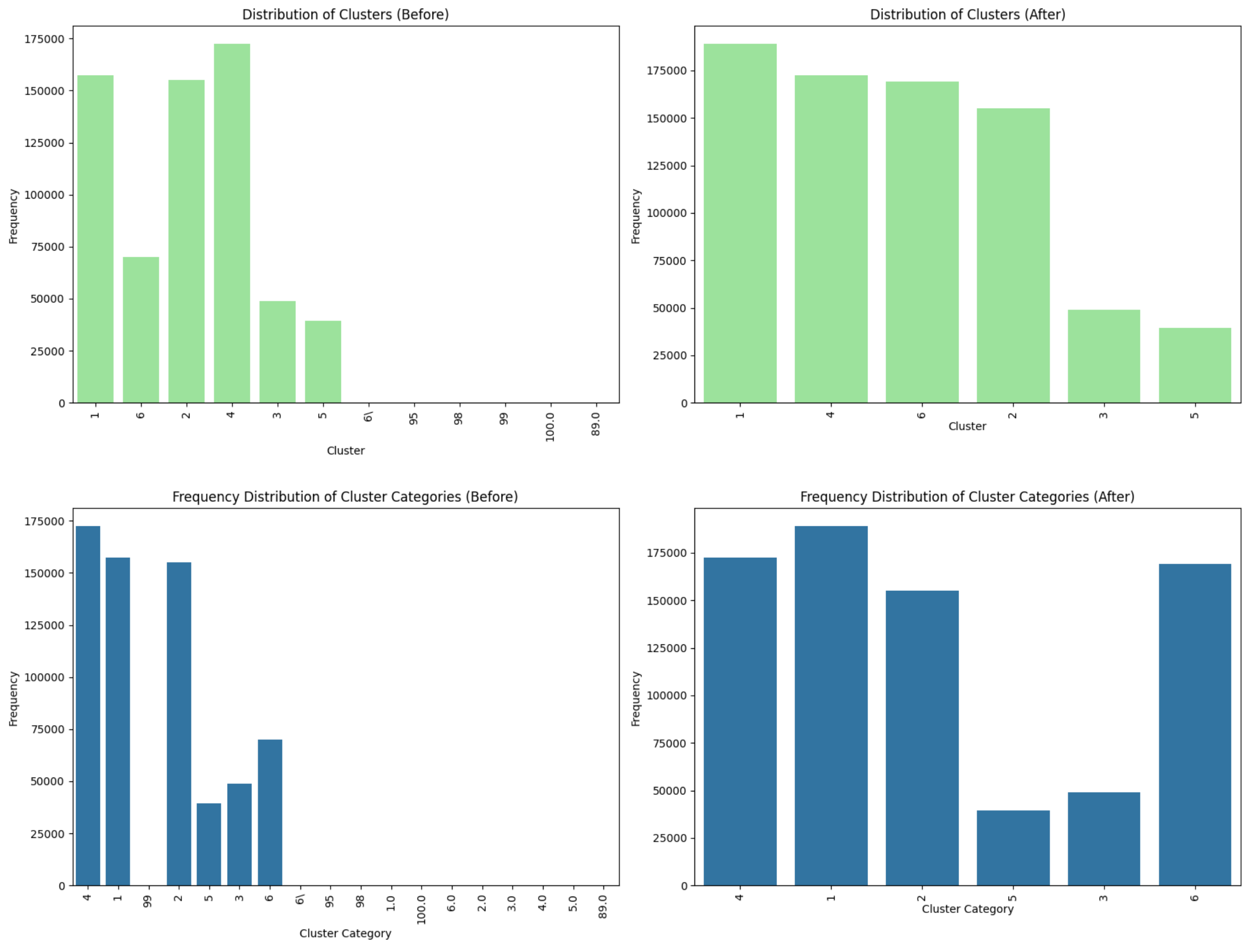
EDA Before & After Data Cleaning - Train CSV



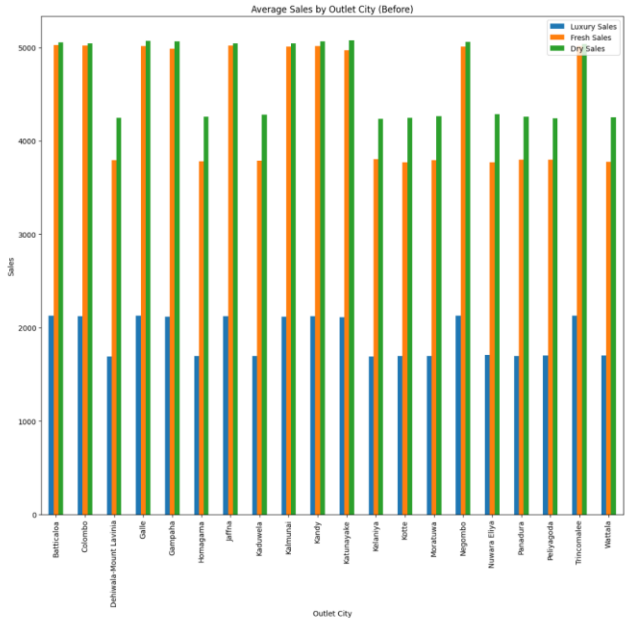
Before: The graph shows variations and inconsistencies in the spelling of city names (e.g., "Batticalse" vs "Batticaloa," "Gulle" vs "Galle," "trincomalee" vs "Trincomalee," "Peyagoda" vs "Peliyagoda," and "Wittala" vs "Wattala").

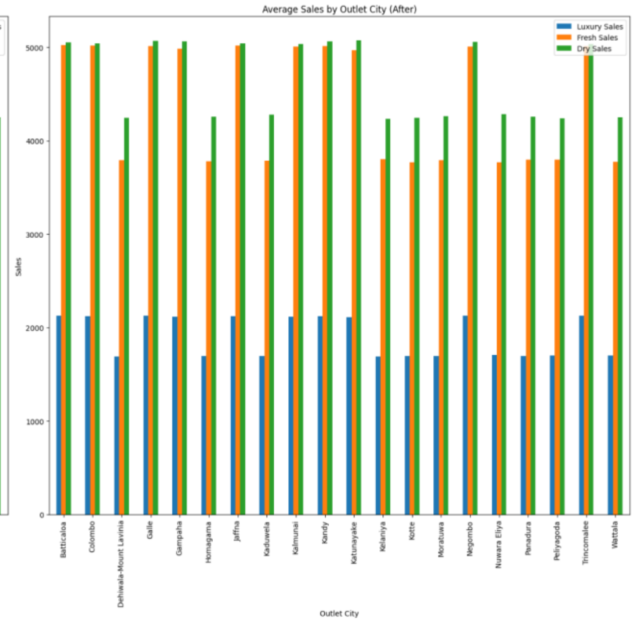
After: The city names are standardised, and the inconsistencies in spelling are corrected.


Distribution of Luxury Sales (Before)
Frequency
Luxury Sales
Distribution of Luxury Sales (After)
Frequency
Luxury Sales
Distribution of Fresh Sales (Before)
Frequency
Fresh Sales
Distribution of Fresh Sales (After)
Frequency
Fresh Sales
Distribution of Dry Sales (Before)
Frequency
Dry Sales
Distribution of Dry Sales (After)
Frequency
Dry Sales



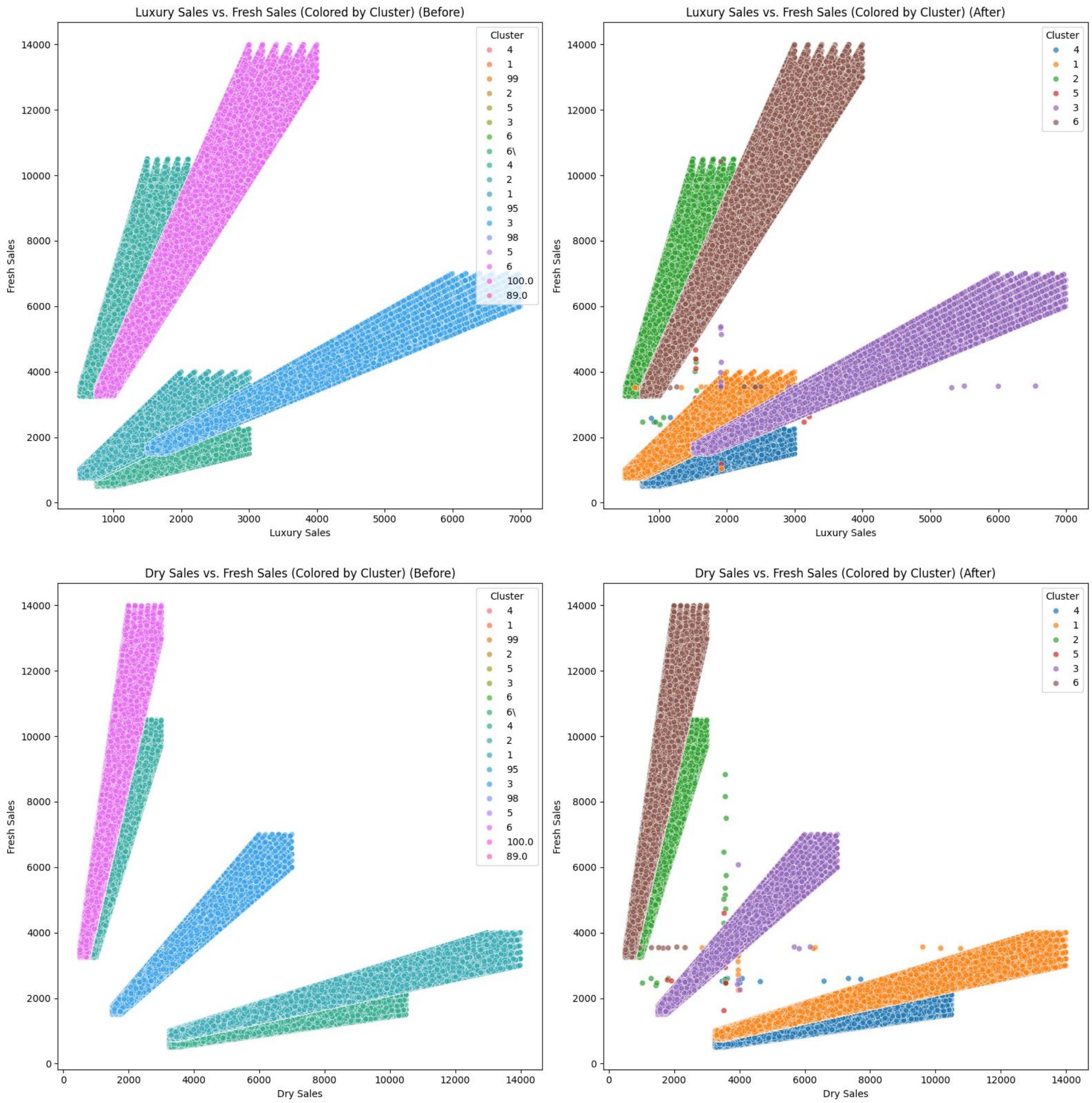
After cleaning, all the distributions changed in terms of the frequency of variation.

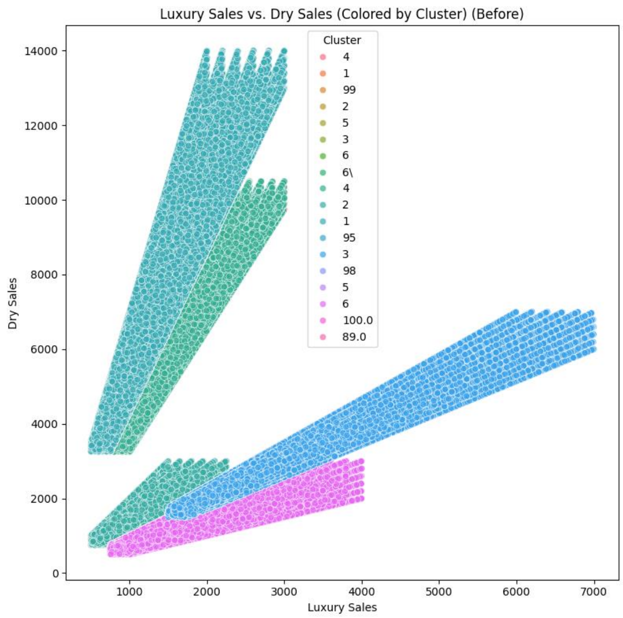


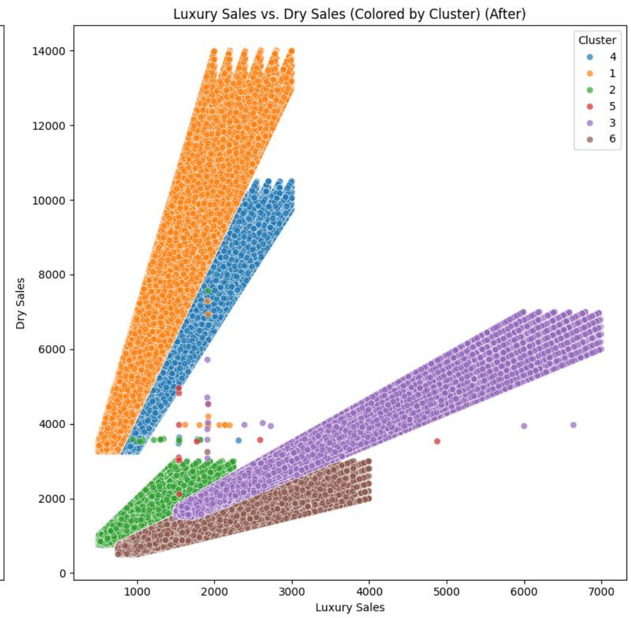


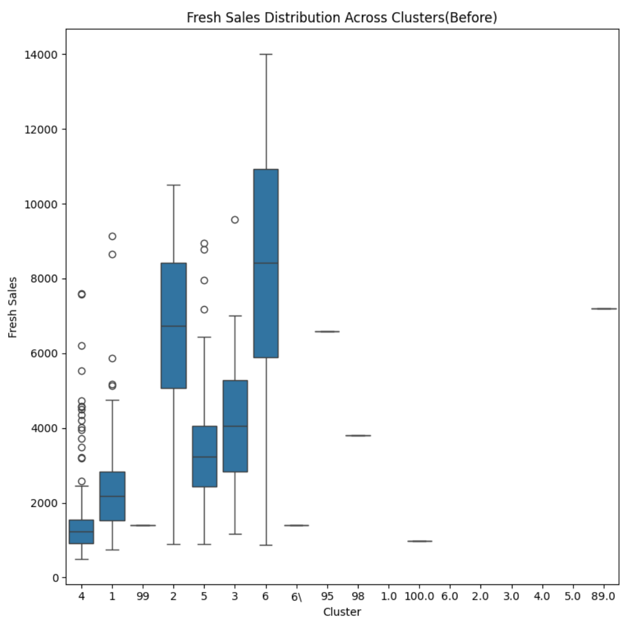
Before: The average sales for different outlet cities are shown with some variations, and city name inconsistencies exist.

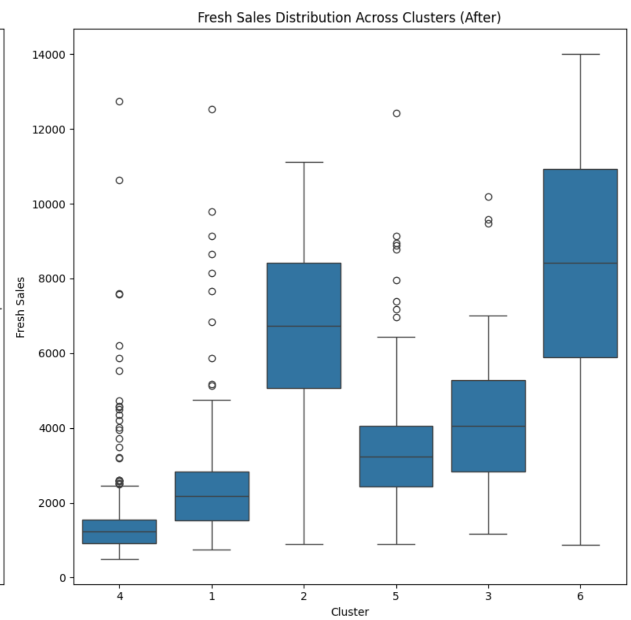
After: The city names are standardised, and the average sales values for some cities have changed.

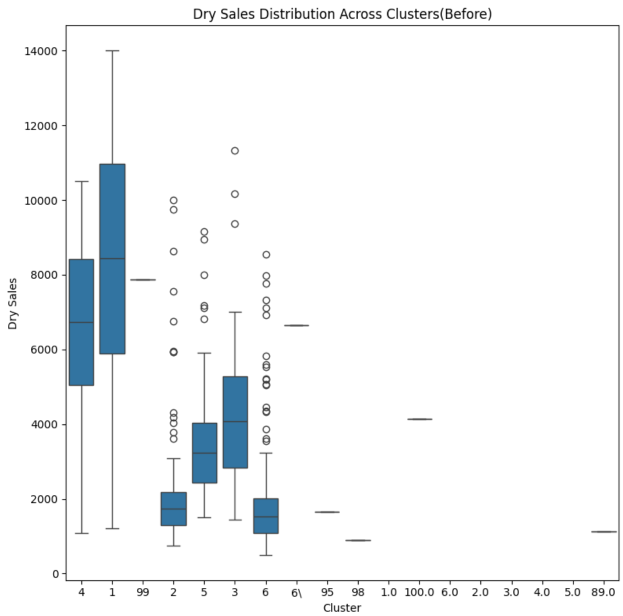

Luxury Sales vs. Fresh Sales (Colored by Cluster) (Before)
Luxury Sales vs. Fresh Sales (Colored by Cluster) (After)
Dry Sales vs. Fresh Sales (Colored by Cluster) (Before)
Dry Sales vs. Fresh Sales (Colored by Cluster) (After)
Fresh Sales
Luxury Sales
Dry Sales
Cluster
4
1
99
2
5
3
6
6\
4
2
1
95
3
98
5
6
100.0
89.0
Cluster
4
1
2
5
3
6

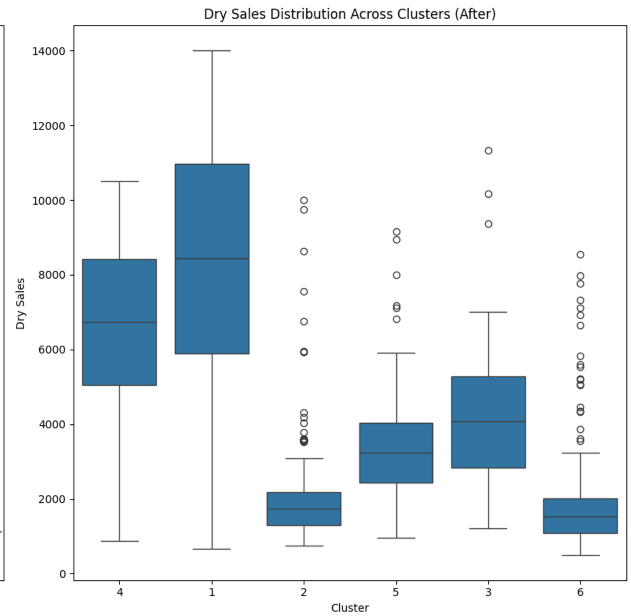

Luxury Sales vs. Dry Sales (Colored by Cluster) (Before)
Cluster
4
1
99
2
5
3
6
6\
4
2
1
95
3
98
5
6
100.0
89.0
Dry Sales
Luxury Sales

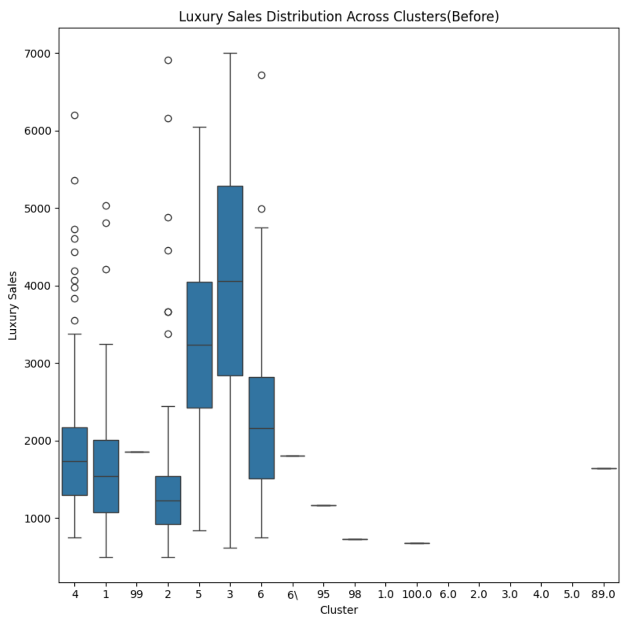

Luxury Sales vs. Dry Sales (Colored by Cluster) (After)
Cluster
4
1
2
5
3
6
Dry Sales
Luxury Sales

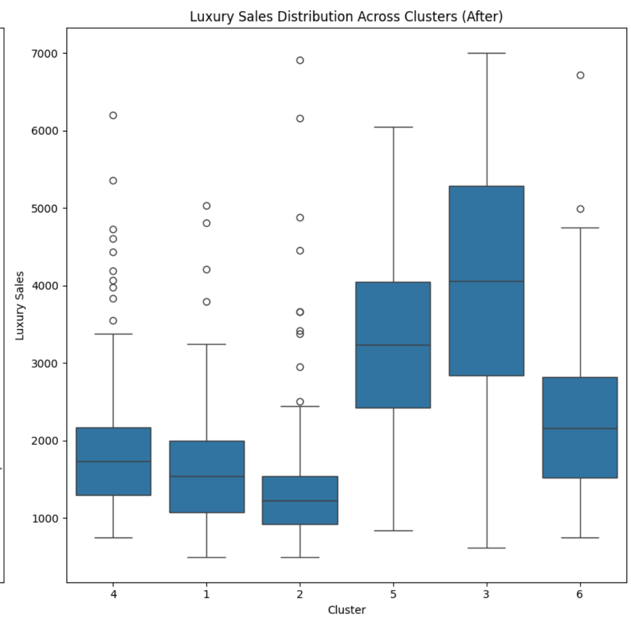

Fresh Sales Distribution Across Clusters(Before)
Fresh Sales
4
1
99
2
5
3
6
6\
95
98
1.0
100.0
6.0
2.0
3.0
4.0
5.0
89.0
Cluster


Fresh Sales Distribution Across Clusters (After)
Fresh Sales
4
1
2
5
3
6
Cluster



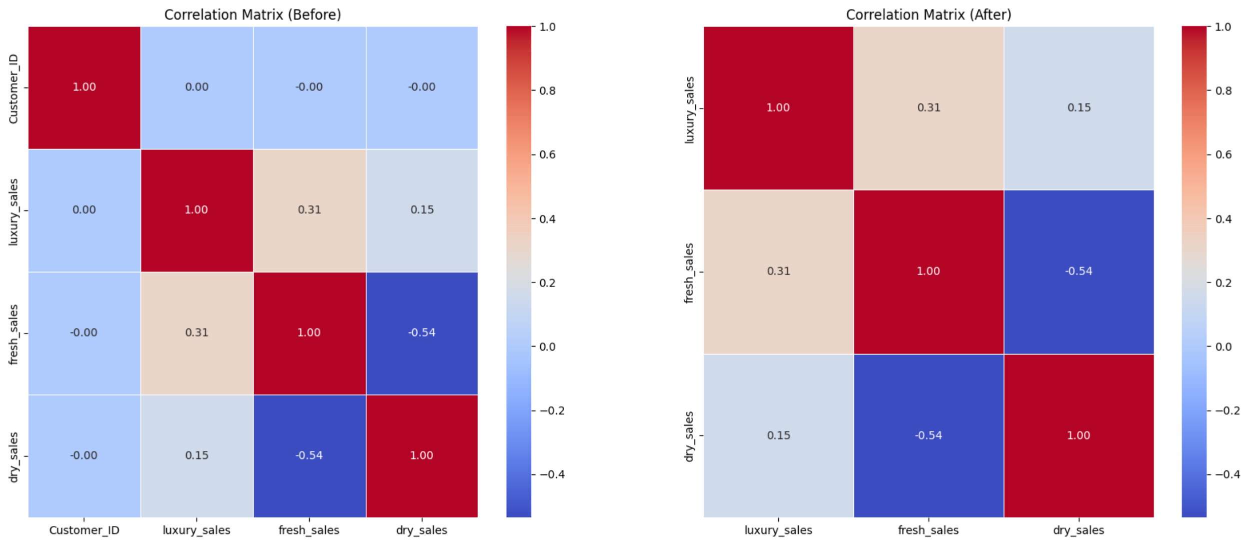






The "Before" graphs show a more scattered distribution of data points.

The "After" graphs show a more refined distribution, suggesting that cleaning has reduced the spread of the data after the removal of invalid clusters. The cluster colouring may also show differences in how the clusters are distributed after cleaning.



The "After" correlation matrix highlights important relationships:

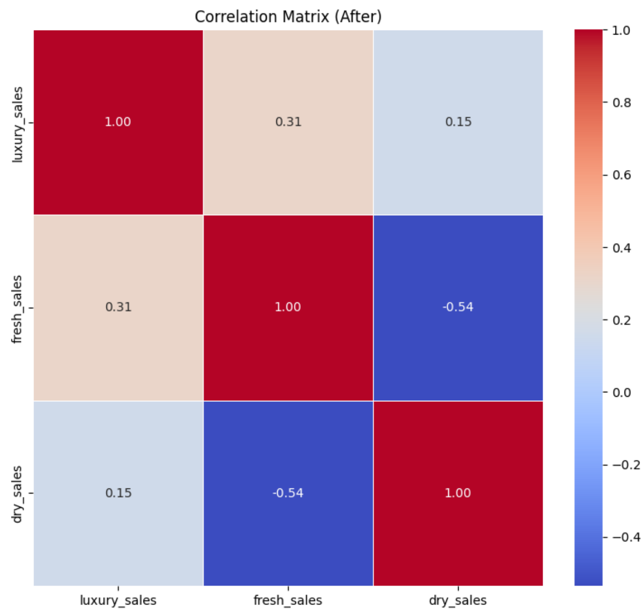
1. There is a strong negative correlation between fresh and dry sales, indicating that customers may favour one category over the other.

2. Luxury sales show a moderate positive correlation with fresh sales but a weak correlation with dry sales, suggesting that luxury purchases are more aligned with fresh food items than with dry goods.

Inter-feature Relationships

The target variable is “cluster_category”. There are a few features that correlate with it. As the cluster category represents information based on the spending patterns of customers across dry sales, luxury sales and fresh sales, the target variable is correlated with these three variables. This is evident through the box plots displayed after cleaning.

The sales variables, luxury sales, dry sales and fresh sales, have strong relationships with each other, as customers who spend in one sales category are more likely to spend in another sales category. This is evident in the scatter plots displayed between sales categories.



Furthermore, by observing the correlation matrix, it is evident that dry sales and fresh sales also have a correlation between them. This can be explained using the type of goods that fall under fresh and dry sales, both of which are usually essential goods that a regular household purchases routinely. Thus, when fresh sales are purchased, dry goods are also simultaneously purchased and vice versa. However, since dry sales contain goods that do not perish very quickly compared to fresh sales, the correlation is not very high, as customers might have to purchase fresh goods more frequently compared to dry goods.

On the other hand, the lack of or the low correlation with luxury sales also can be attributed to the type of products in this category, as luxury products are not purchased frequently by customers and are more often than not purchased separately from fresh and dry goods.

The variable “outlet_city” seems to have minimal correlation with other variables, as it doesn't seem to have an effect on sales variables. This is evident through the bar charts displayed after cleaning.

Target Variable Analysis

The dataset consists of 6 classified clusters; below are appropriate names for each cluster based on distinctive customer spending patterns:

- Segment 1 - Practical bulk buyers:

  This segment is highly focused on dry sales with moderate fresh and luxury spending sales. They are likely to be bulk buyers of dry goods. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
1,075	2,004	1,520	2,824	5,892	10,970

- Segment 2 - Fresh-only buyers:

  This segment is highly focused on fresh sales. Dry sales and Luxury sales are significantly low compared to fresh sales. Customers are likely to focus on perishable goods. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
1,544	9,525	5,058	8,429	1,302	2,171

- Segment 3 - Premium All-round buyers:

  This segment consists of balanced spending of high values across all segments. They may tend to be premium, generalised customers. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
2,845	5,289	2,842	5,282	2,840	5,280

- Segment 4 - Necessity bulk buyers:

  This segment is highly dominant in the dry sales segment, while luxury sales and fresh sales tend to be significantly low in comparison. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
1,299	2,172	922	1,547	5,043	8,427

- Segment 5 - All-Round buyers:

  This segment also consists of balanced spending across all segments. However, the spending amounts aren't as high as in segment 3. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
2,428	4,053	2,426	4,053	2,427	4,045

- Segment 6 - Selective fresh buyers:

  This segment is highly focused on fresh sales, while there is moderate spending in luxury sales, while dry sales are the lowest segment. This segment has the following upper and lower bounds for each sales category:

Luxury Sales		Fresh Sales		Dry Sales	
Lower Bound	Upper Bound	Lower Bound	Upper Bound	Lower Bound	Upper Bound
1,521	2,818	5,900	10,936	1,078	2,001

Modelling Approach

Model Selection

Having completed the project's Data Cleaning and EDA parts, creating the machine learning model for customer segmentation is now possible. Since the machine learning model will be trained using already labelled data, which is the customer segment column in the train data, supervised machine learning models must be used. Furthermore, since the target variable – customer segmentation- is a categorical variable with six categories, the ML model must also be a classification model.

Logistic Regression

One option that meets these criteria is a logistic regression model. Logistic regression is a classification algorithm that predicts the probability of a customer belonging to one out of several segments using a logistic function. The utility of this model is quite good, especially for problems of binary classification, where a derived metric (out of two possible classes) is a likely outcome.

Datasets such as customer attributes such as demographic information, purchasing behaviour, etc., can be worked on to find segments using logistic regression. Thus, it allows customers to be laid down into different segments so that they are marketed in a more focused manner. Logistic regression has been used in a study done at Strathmore University where the study used it to segment customers in the telecommunications sector. According to Omonge (2021), the model was intended to help the company cluster its customers based on their purchasing patterns so that it can provide personalized services to them.

KNN

KNN is a classification algorithm that classifies new customers by finding the K closest data point (Neighbors) in the train data set. In a KNN model, neighbours are categorised in a certain way; you will likely fall under the same category, too, based on the virtue of you being near to them.

Conversely, those who are near but farther away are likely to fall under a different classification than yours. (Sharma, 2022).

KNN is used to segment customers based on their purchasing behaviour. KNN can easily group customers into segments with similar choices and preferences. This helps businesses target the right customers with the right products or advertisements. (Geeksforgeeks, 2025) One of our tasks was to label the test dataset, and with an unlabelled dataset, we could cluster the customers using the K-Means algorithm. Hence, we generate a label based on its cluster result. (RPUBS - Customer segmentation, 2020)

Decision Tree

A Decision Tree is a rule-based model that splits data based on feature values, forming a tree-like structure. A decision tree begins with a root node and then splits into similar groups called decision nodes. Each node represents a customer feature. The lines that connect the nodes are identified as branches. These are decision rules that help divide customers into groups or segments.

For predictive modelling, decision trees provide a structured way to visualise the results of each decision node and a very flexible approach to creating that structure. By inputting sales data in luxury, dry fresh and outlet cities, we will be able to predict the range of consumer spending patterns in different outlet cities.

This can help marketing strategies as they target the different marketing campaigns for the right segments. By incorporating decision tree models in a business setting, the organisation can adopt a sophisticated analytical tool that can immensely help in decision-making. (Understanding decision trees, no date)

Random Forest

A Random Forest is an ensemble machine-learning method that builds several decision trees and combines their output to predict the target variable. Each tree in the forest is trained on a separate subset of the data and features, which helps reduce the model's overfitting and better generalise. In such a classification case, the final output will be decided by majority voting (Gradient, 2016).

This is a good model for such tasks as it ensures that the segments are created with minimal bias. Therefore, greater accuracy can be obtained, thus making confident decisions on these insights (Abdullahi, A, Htet, S, et al., 2023).

Gradient Boosting

Gradient Boosting is an ensemble learning method that creates decision trees sequentially, with each tree improving on the errors made by the previous trees. Then, the target variable is given by the combined output of all these trees, which gives a highly accurate model that is sensitive to even small trends and patterns in the data. While this model can be a more computationally intensive model, once it is fine-tuned, the obtained predictions can help identify each segment accurately and use more targeted marketing strategies.

There is evidence in previous studies, such as that of Kabir, M (2025), that a gradient-boosting model outperformed other models when it comes to customer segmentation. Similarly, Venkatakrishna R (2021) further shows how a Gradient Boosting model performs well at the task of customer segmentation.

Baseline Models

Baseline models are created for each of these algorithms so that the effect of data scaling and feature engineering can be evaluated by comparing the baseline models with the models made after these techniques have been implemented.

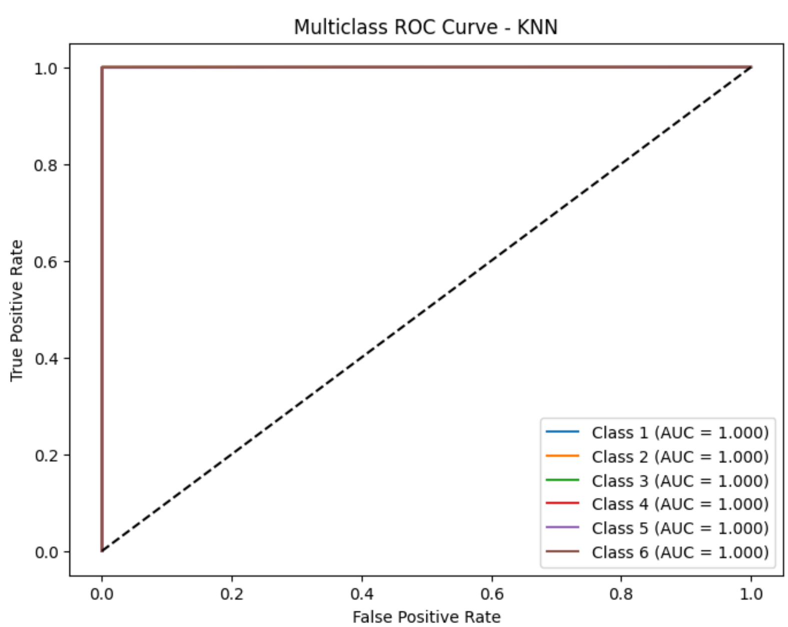
Model Training & Hyperparameter Tuning

Once the feature scaling has been performed on the dataset, the previous five models can be trained again using the now-scaled data. This should provide better and more accurate models. Moreover, in order to further enhance these models, the best hyperparameters need to be selected for these models, which is referred to as hyperparameter tuning.

KNN

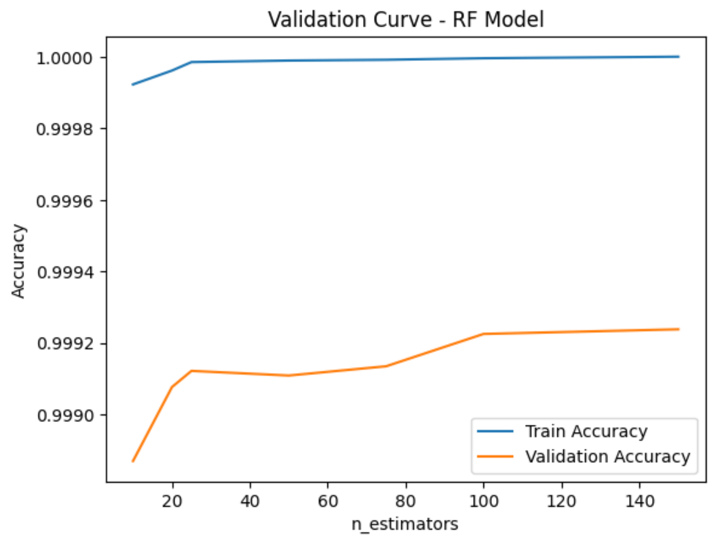
In the case of the KNN model, the hyperparameter that is being tuned is the K value, which happens to be the number of neighbours. In order to obtain the best K value, the tuning technique employed is a search with cross-validation, which is a highly effective way to obtain the optimum k value, with the only drawback being it requires somewhat high computational power. In this case, the grid search for the KNN model was run with a range from 1 to 10, and the optimum k value was obtained at 5.

Furthermore, the ROC and the ROC-AUC value were used to check the class distinguishing ability of the KNN model and obtained an AUC value of 0.99985 and the following graph:



Random Forest

In the Random Forest model, the hyperparameter being tuned is the n_estimators parameter, which is the number of decision trees created inside the Random Forest. In order to obtain the optimum value for this hyperparameter, similar to the case with the KNN model, a grid search with cross-validation was used in addition to a train accuracy and validation accuracy graph. The values used for the grid search were 10, 15, 20, 25, 50, 75 and 100, and the following graph was obtained.



Based on the interpretation from the graph and the grid search, it was concluded that the n_estimators hyperparameter started having diminishing returns at the 25 mark, meaning that further increase in the hyperparameter, while it would slightly increase the model accuracy, could not be enough to be a significant impact for the additional computational power required to run the model at higher n_estimators to be justified.

Model Evaluation

In order to evaluate the models that have been built and to select the most suitable one, each model was subject to several evaluation techniques. First and foremost, for each model, the accuracy score, MSE and R^2 values are calculated to compare against each other to see the best-performing model. The accuracy score is the correct prediction out of the total predictions made by the model, and the MSE gives the mean squared difference between the actual values and the predicted values. The r2 value tells how much of the variance in the target variable is explained by the model.

Model	Accuracy Score	MSE	R2
Logistic Regression (Baseline)	0.9636	0.1702	0.9513
KNN (Baseline)	0.9526	0.2163	0.9382
Decision Tree (Baseline)	0.9526	0.2163	0.9382
Random Forest (Baseline)	0.9993	0.0062	0.9982
Gradient Boosting (Baseline)	0.9766	0.1392	0.9602
Logistic Regression	0.9628	0.1801	0.9484
KNN	0.9998	0.0012	0.9997
Decision Tree	0.9982	0.0187	0.9946
Random Forest	0.9992	0.0084	0.9992
Gradient Boosting	0.9746	0.1473	0.9578

While these measures provide a good understanding of the models, there are other criteria to consider as well, which include recall and precision scores as well as checking for under or overfitting. The recall and precision scores for each model can be obtained through their relevant classification reports.

KNN Model

- Accuracy: 99.98%

It is the percentage of correct predictions from total predictions. This means that 99.98% of the predictions made by the KNN model were accurate.

- MSE: 0.0012

It is the mean squared difference between the actual values and the predicted value. The very low value of 0.0012 suggests that there is a very low distance between the actual and predicted values in the KNN model.

- R2: 0.9997

The r2 value indicates how much of the variance is explained by the model. In this case, the model explains 99.97% of the variance.

	precision	recall	f1-score	support
1	1.00	1.00	1.00	37876
2	1.00	1.00	1.00	30922
3	1.00	1.00	1.00	9748
4	1.00	1.00	1.00	34370
5	1.00	1.00	1.00	7962
6	1.00	1.00	1.00	33951
accuracy			1.00	154829
macro avg	1.00	1.00	1.00	154829
weighted avg	1.00	1.00	1.00	154829

- Precision: 1.00 for all six classes

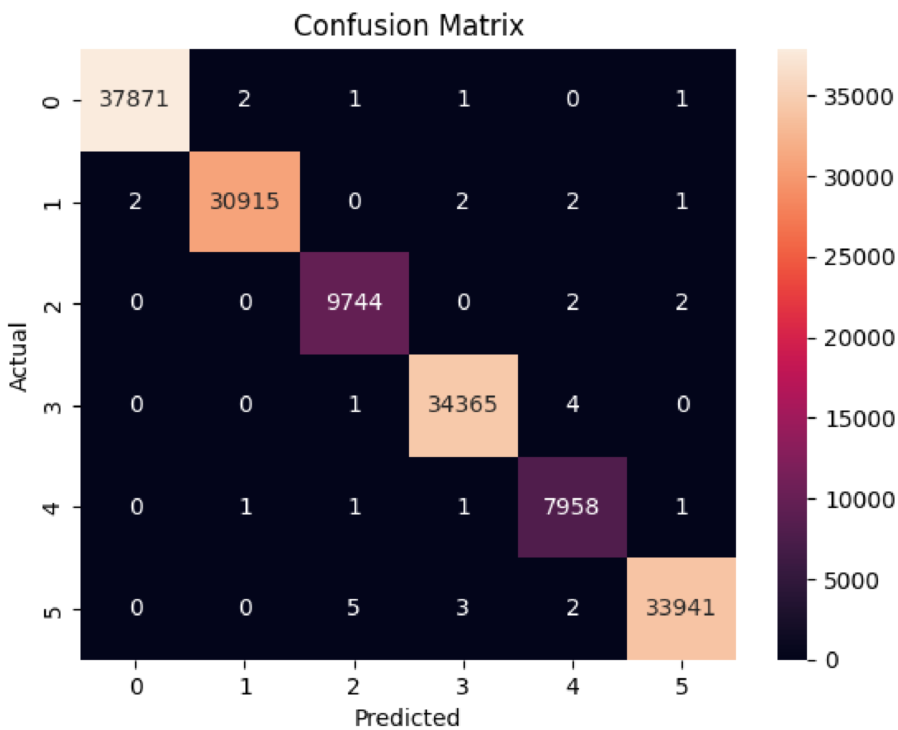
This is a measure of how many of the positive predictions are actually correct. A 1.00 precision value for all the classes means that the KNN model has made perfect positive predictions for all six classes.

- Recall: 1.00 for all six classes

This measure is also similar to precision; it measures how many of the actual positives were correctly identified. A recall value of 1.00 for all six classes indicates that the KNN model nearly perfectly identified all the actual positive values.

- Confusion Matrix:

It is a presentation of the actual and predicted values obtained using a model. It displays how many of the predictions are correct and incorrect for each class.



- Overfitting Check:

Training Accuracy: 0.9997, Validation Accuracy: 0.9998, Test Accuracy: 0.9998

These values were compared to check if the KNN model was overfitting or underfitting. Since the validation accuracy and test accuracy are the same and both are greater than the training accuracy, thus it was concluded that the model was well-fitted.

- Class distinguishability: ROC-AUC - 0.99985

The ROC-AUC value was calculated to check for the class distinguishability of the KNN model. The very high value of 0.99985 means that the KNN model has excellent discrimination ability between the six classes of the cluster category.

Logistic Regression Model

- Accuracy: 96.28%

It is the percentage of correct predictions from total predictions. This means that 96.28% of the predictions made by the logistic regression model were accurate.

- MSE: 0.1801

It is the mean squared difference between the actual values and the predicted value. The low value of 0.1801 suggests that there is a low distance between the actual and predicted values in the KNN model.

- R2: 0.9484

The r2 value indicates how much of the variance is explained by the model. In this case, the model explains 94.84% per cent of the variance.

	precision	recall	f1-score	support
1	1.00	1.00	1.00	37876
2	1.00	1.00	1.00	30922
3	0.72	0.76	0.74	9748
4	1.00	1.00	1.00	34370
5	0.69	0.64	0.66	7962
6	1.00	1.00	1.00	33951
accuracy			0.96	154829
macro avg	0.90	0.90	0.90	154829
weighted avg	0.96	0.96	0.96	154829

- Precision: clusters 3 and 5 at 0.72 and 0.69, rest at 1.00

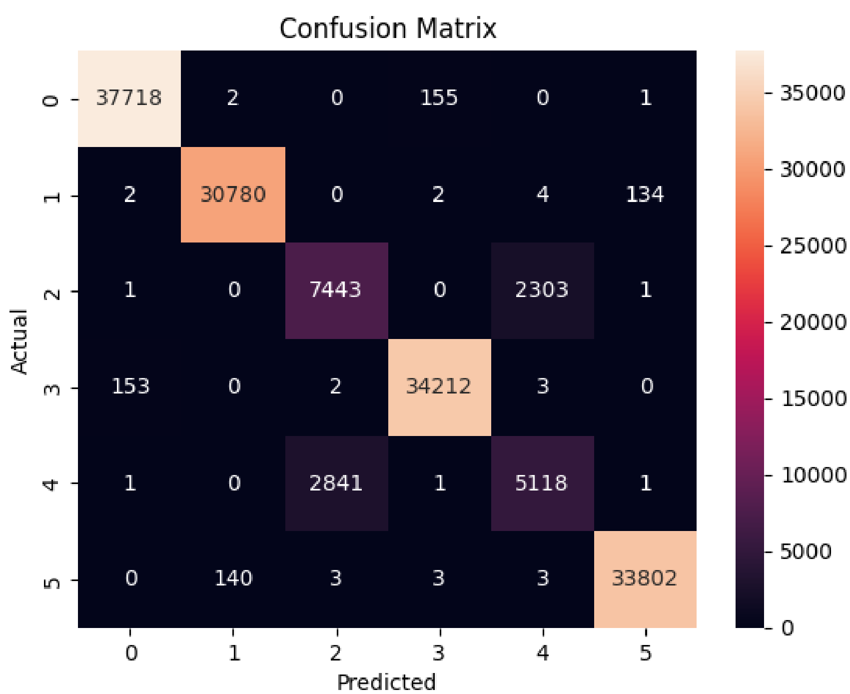
This is a measure of how many of the positive predictions are actually correct. A 1.00 precision value for the classes means that the model has made near-perfect positive predictions for these classes. However, when it comes to clusters 3 and 5, only about 70% were accurate predictions.

- Recall: clusters 3 and 5 at 0.74 and 0.66, rest at 1.00

This measure is also similar to precision; it measures how many of the actual positives were correctly identified. A recall value of 1.00 for the classes indicates that the model nearly perfectly identified all the actual positive values. However, for clusters 3 and 5, only about 70% of the actual positives were captured.

- Confusion Matrix:

It is a presentation of the actual and predicted values obtained using a model. It displays how many of the predictions are correct and incorrect for each class.

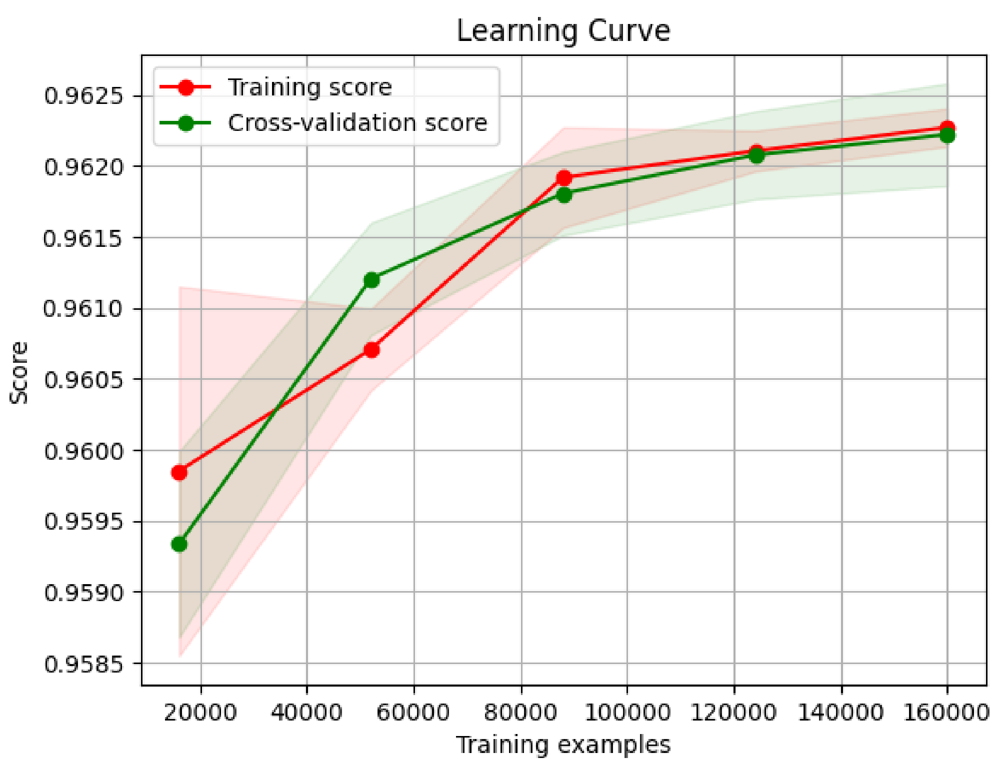


- Overfitting Check:

Training Accuracy: 0.9629, Validation Accuracy: 0.9623, Test Accuracy: 0.9628

These values were compared to check if the logistic regression model was overfitting or underfitting. Since the validation accuracy and test accuracy are the same and both are greater than the training accuracy, thus it was concluded that the model was well-fitted.

In addition to this, a learning curve was also used to observe how this model would improve as the amount of training data and iterations is increased. This curve can also be interpreted to check for the over and underfitting of the model. As can be observed, since the training curve and the validation curve are high and close together, it can be concluded that this model is well-fitted.



Decision Tree Model

- Accuracy: 99.83%

It is the percentage of correct predictions from total predictions. This means that 99.83% of the predictions made by the decision tree model were accurate.

- MSE: 0.0167

It is the mean squared difference between the actual values and the predicted value. The very low value of 0.0167 suggests that there is a very low distance between the actual and predicted values in the decision tree model.

- R2: 0.9952

The r2 value indicates how much of the variance is explained by the model. In this case, the model explains 99.52% per cent of the variance.

	precision	recall	f1-score	support
1	1.00	1.00	1.00	37876
2	1.00	1.00	1.00	30922
3	1.00	1.00	1.00	9748
4	1.00	1.00	1.00	34370
5	1.00	1.00	1.00	7962
6	1.00	1.00	1.00	33951
accuracy			1.00	154829
macro avg	1.00	1.00	1.00	154829
weighted avg	1.00	1.00	1.00	154829

- Precision: 1.00 for all six classes

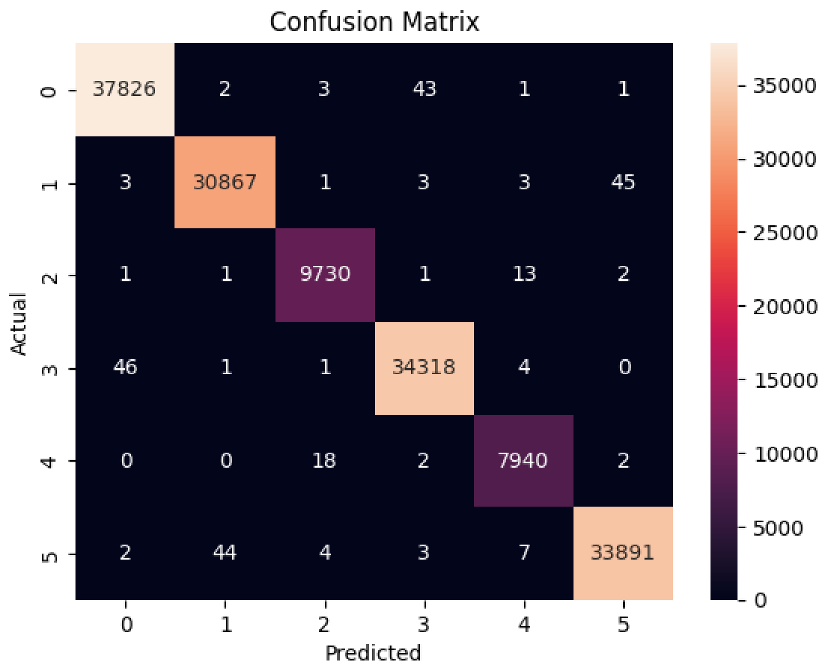
This is a measure of how many of the positive predictions are actually correct. A 1.00 precision value for all the classes means that the decision tree model has made a near-perfect positive prediction for all six classes.

- Recall: 1.00 for all six classes

This measure is also similar to precision; it measures how many of the actual positives were correctly identified. A recall value of 1.00 for all six classes indicates that the Decision tree model nearly ideally identified all the actual positive values.

- Confusion Matrix:

It is a presentation of the actual and predicted values obtained using a model. It displays how many of the predictions are correct and incorrect for each class.



- Overfitting Check:

Training Accuracy: 1.0000, Validation Accuracy: 0.9983, Test Accuracy: 0.9983

These values were compared to check if the logistic regression model was overfitting or underfitting. Since the validation accuracy and test accuracy are the same and both are greater than the training accuracy, thus it was concluded that the model was well-fitted.

Random Forest Model

- Accuracy: 99.93%

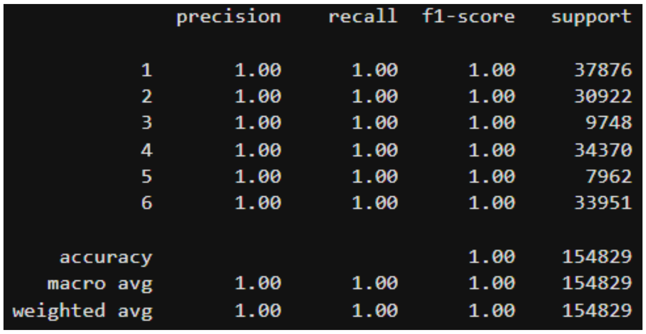
It is the percentage of correct predictions from total predictions, which means that 99.93% of the predictions made by the Random Forest model were accurate.

- MSE::0.0064

It is the mean squared difference between the actual values and the predicted value. The very low value of 0.0064 suggests that there is a very low distance between the actual and predicted values in the Random forest model.

- R2: 0.9982

The r2 value indicates how much of the variance is explained by the model. In this case, the model explains 99.82% per cent of the variance.



	precision	recall	f1-score	support
1	1.00	1.00	1.00	37876
2	1.00	1.00	1.00	30922
3	1.00	1.00	1.00	9748
4	1.00	1.00	1.00	34370
5	1.00	1.00	1.00	7962
6	1.00	1.00	1.00	33951
accuracy			1.00	154829
macro avg	1.00	1.00	1.00	154829
weighted avg	1.00	1.00	1.00	154829

- Precision: 1.00 for all six classes

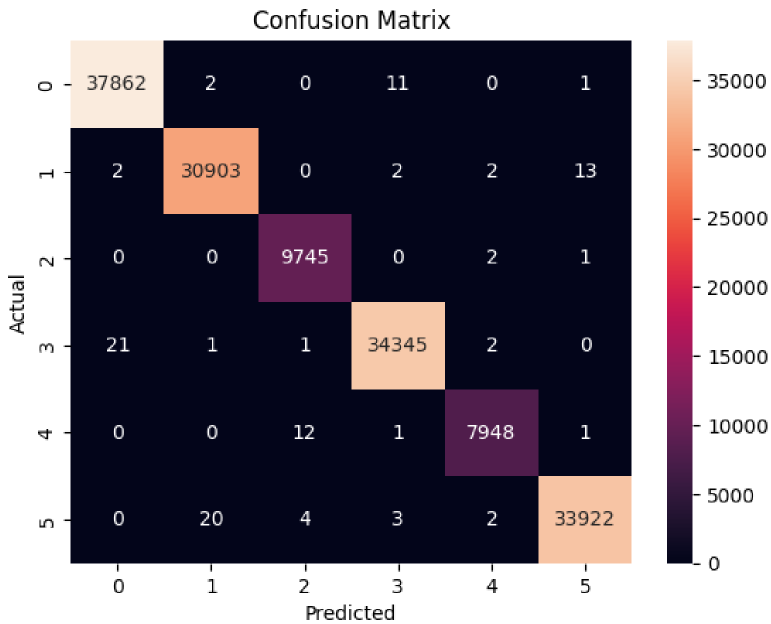
This is a measure of how many of the positive predictions are actually correct. A 1.00 precision value for all the classes means that the Random Forest model has made near-perfect positive predictions for all six classes.

- Recall: 1.00 for all six classes

This measure is also similar to precision; it measures how many of the actual positives were correctly identified. A recall value of 1.00 for all six classes indicates that the Random Forest model nearly ideally identified all the actual positive values.

- Confusion Matrix:

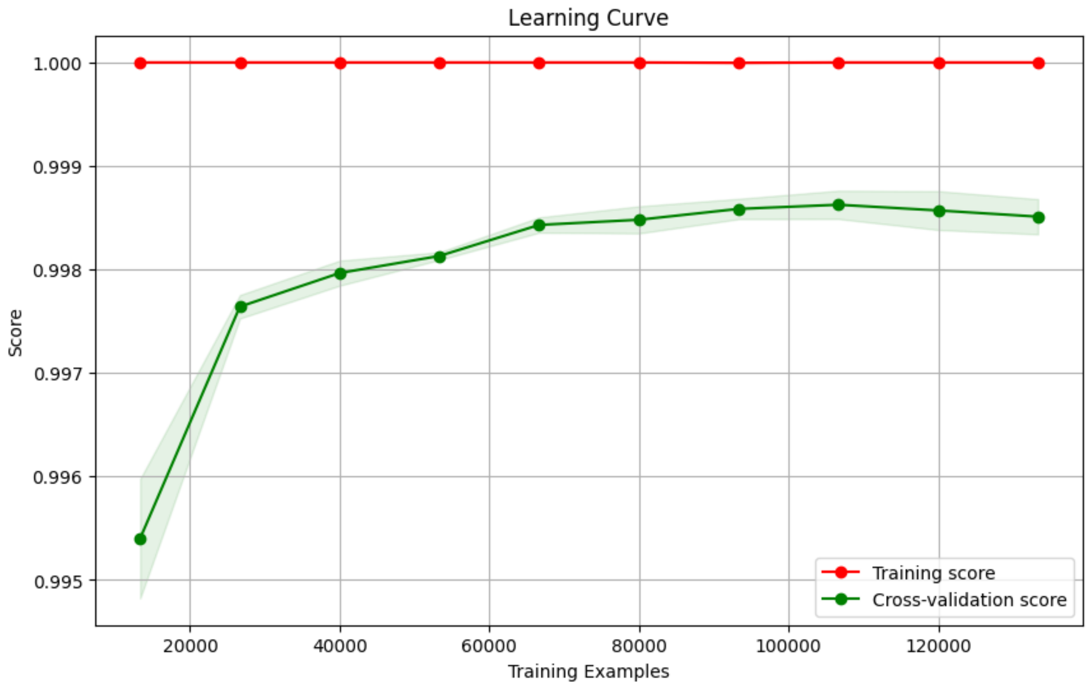
It is a presentation of the actual and predicted values obtained using a model. It displays how many of the predictions are correct and incorrect for each class.



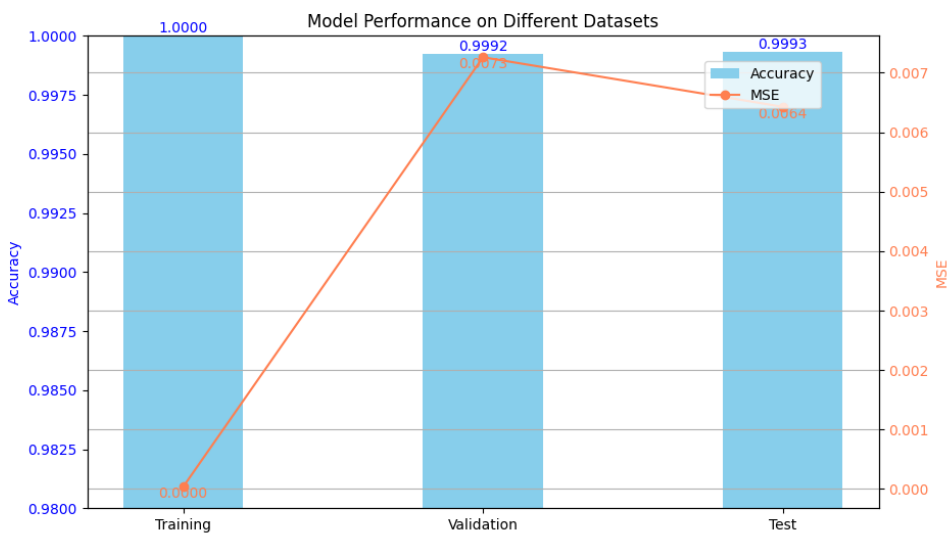
- Overfitting Check:

Training Accuracy: 1.0000, Validation Accuracy: 0.9992, Test Accuracy: 0.9993 These values were compared to check if the logistic regression model was over or underfitting. Since the validation accuracy and test accuracy are the same and both are greater than the training accuracy, thus it was concluded that the model was well-fitted.

In addition to this, a learning curve was also used to observe how this model would improve as the amount of training data and iterations is increased. This curve can also be interpreted to check for the over and underfitting of the model. As can be observed, since the training curve and the validation curve are high and close together, it can be concluded that this model is well-fitted.



The following graph can be used to reinforce the same point as it shows the accuracy levels of the model across different subsets of the dataset.



Gradient Boosting Model

- Accuracy: 97.46%

It is the percentage of correct predictions from total predictions. This means that 97.46% of the predictions made by the Gradient Boosting model were accurate.

- MSE::0.1473

It is the mean squared difference between the actual values and the predicted value. The low value of 0.1473 suggests that there is a low distance between the actual and predicted values in the gradient-boosting model.

- R2: 0.9578

The r2 value indicates how much of the variance is explained by the model. In this case, the model explains 95.78% per cent of the variance.

	precision	recall	f1-score	support
1	0.99	0.99	0.99	37876
2	0.99	1.00	0.99	30922
3	0.84	0.85	0.85	9748
4	0.99	0.99	0.99	34370
5	0.81	0.80	0.81	7962
6	1.00	0.99	0.99	33951
accuracy			0.97	154829
macro avg	0.94	0.94	0.94	154829
weighted avg	0.97	0.97	0.97	154829

- Precision: In the range of 0.80 to 1.00 for all classes

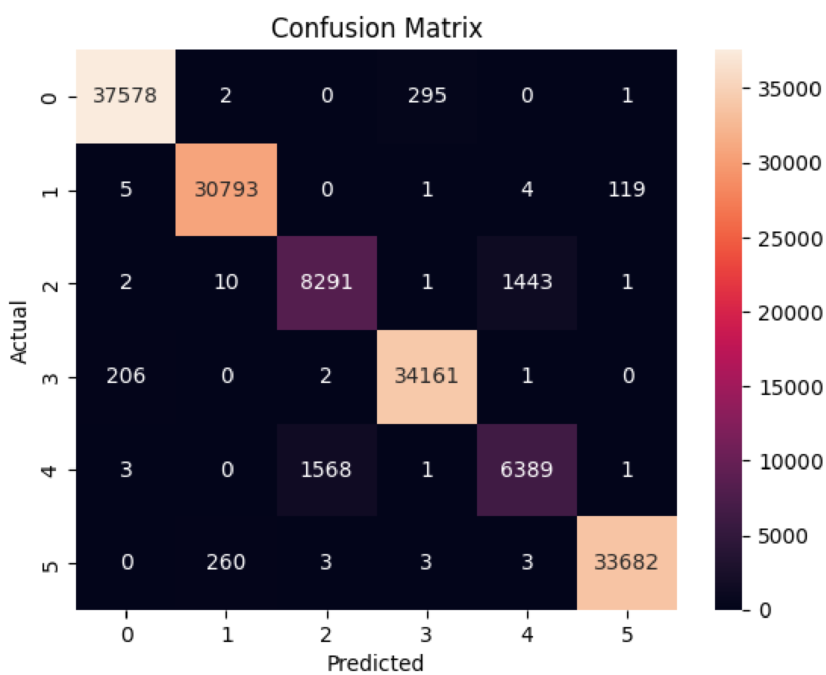
This is a measure of how many of the positive predictions are actually correct. A 1.00 precision value for the classes means that the model has made a near-perfect positive prediction for these classes. However, when it comes to this Model, except for cluster 6, all the other clusters had slightly lower values than 1.00; this does not mean that they performed badly since most of them still had above 90% accuracy. Only clusters 3 and 5 had lower than 90%.

- Recall: In the range of 0.80 to 1.00 for all classes

This measure is also similar to precision; it measures how many of the actual positives were correctly identified. A recall value of 1.00 for the classes indicates that the model nearly perfectly identified all the actual positive values. However, when it comes to this Model, except for cluster 6, all the other clusters had slightly lower values than 1.00; this does not mean that they performed badly since most of them still had above 90% accuracy. Only clusters 3 and 5 had lower than 90%.

- Confusion Matrix:

It is a presentation of the actual and predicted values obtained using a model. It displays how many of the predictions are correct and incorrect for each class.



- Overfitting Check:

Training Accuracy: 0.9748, Validation Accuracy: 0.9738, Test Accuracy: 0.9746 These values were compared to check if the logistic regression model was over or under-fitting. Since the validation accuracy and test accuracy are the same and both are greater than the training accuracy, thus it was concluded that the model was well-fitted.

The accuracy scores of train, test and validation data subsets have been compared to check if the model is overfitting or underfitting. Furthermore, learning and validation curves have also been plotted for some of the models to evaluate them better.

★ The reason for these precision and recall values for clusters 3 and 5 could be explained by their characteristics. The customers in both these clusters are well-balanced customers who purchase from all three types of products, with the only difference being that the customers in cluster 3 purchase higher amounts compared to those of cluster 5. Therefore, the Gradient Boosting Model might have run into issues with this classification and thus split the customers into the wrong customer segment.

Final Model Selected

Final Model: Random Forest

After evaluating multiple classification models, Random Forest was selected as the final model for customer segmentation. This decision was based on its high performance across multiple metrics, ability to handle complex datasets, and interpretability through feature importance analysis.

- <u>**Model Performance**</u>

The Random Forest model achieved a high accuracy and r2 value of 0.9992, as well as a very low MSE value of 0.0084, second only to the KNN model. This high accuracy indicates the model's ability to classify customer segments with minimal errors correctly. In addition, the model demonstrated near-perfect recall and precision scores, suggesting that it can accurately identify all customer categories while minimising false positives and false negatives.

	precision	recall	f1-score	support
1	1.00	1.00	1.00	37876
2	1.00	1.00	1.00	30922
3	1.00	1.00	1.00	9748
4	1.00	1.00	1.00	34370
5	1.00	1.00	1.00	7962
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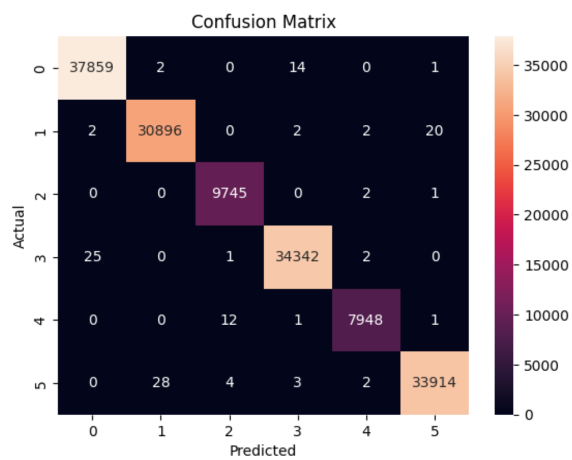
weighted avg	1.00	1.00	1.00	154829



- <u>**Suitability for Complex and Large Datasets**</u>

Models such as KNN use lazy learning. Therefore, it requires a lot of computational resources and time to make predictions. Therefore, it is unsuitable for this problem. On the other hand, Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the class mode for classification. This approach allows the model to handle large and complex datasets effectively, resulting in this being selected over a KNN model.

- **Capturing Non-Linear Patterns**

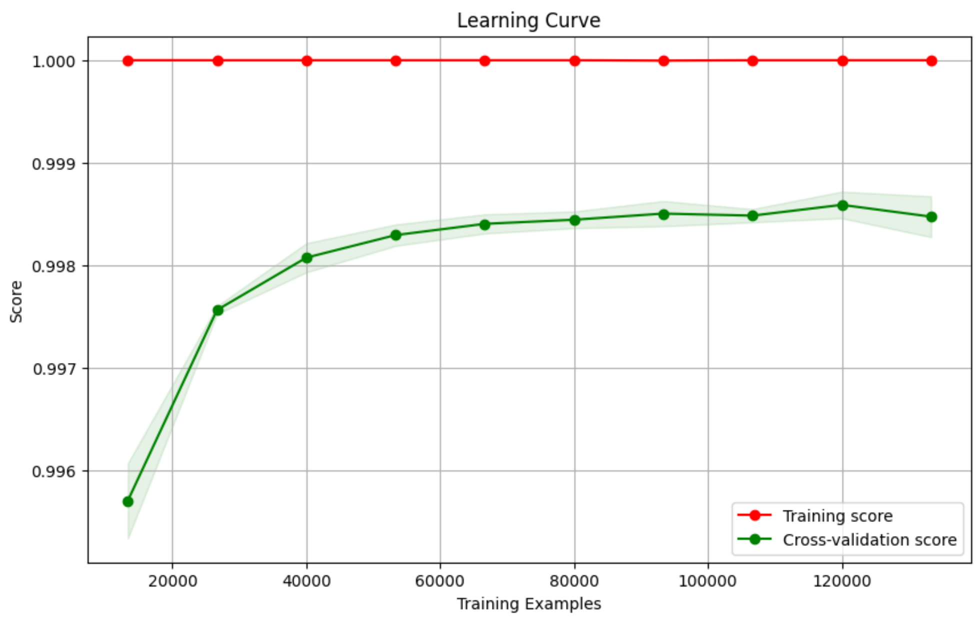
Random Forest models are not limited to linear patterns, such as the case with logistic regression models. Therefore, the random forest model can capture nonlinear patterns and trends in complex datasets.

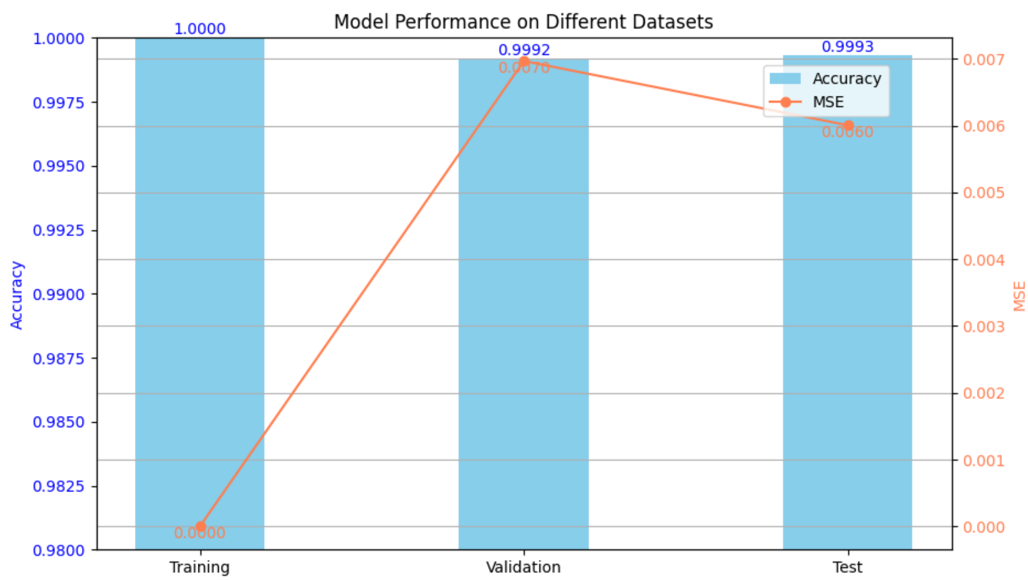
- **Computational Efficiency**

While training Random Forest models can be computationally intensive during training, they are significantly more efficient during prediction compared to KNN, as a random forest model learns during its training and makes future predictions based on that. Therefore, Random Forest is a more practical option for deploying the model in real-world applications requiring rapid predictions.

- **Overfitting and Model Generalization**

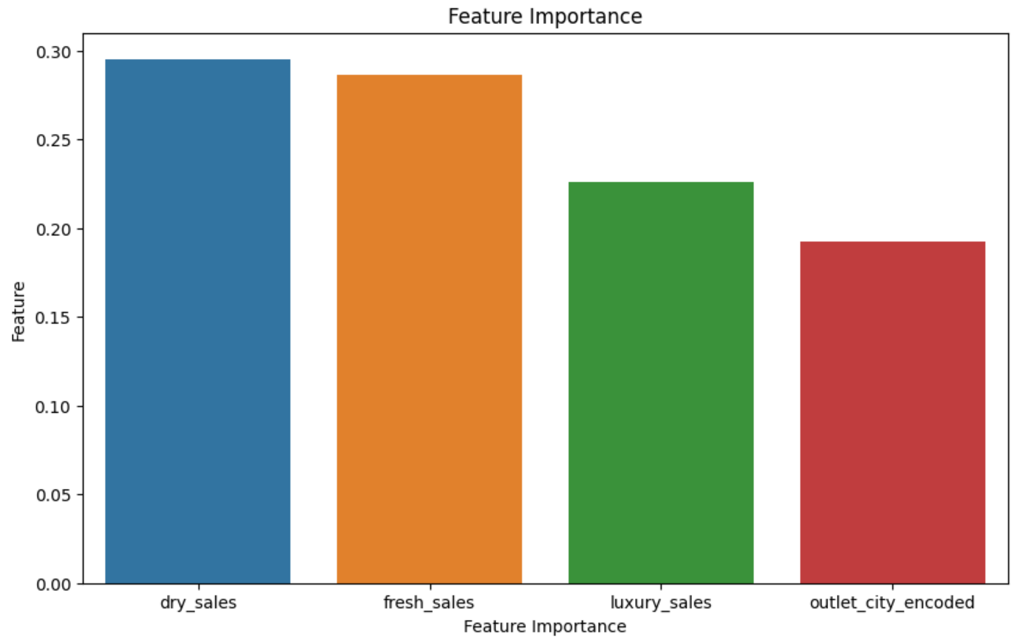
The model's performance on the training, validation, and test datasets shows that it is well-fitted. The fact that accuracy remains consistently high across all datasets indicates that the model generalises well to unseen data without overfitting. Which is reinforced by the learning curve.





- <u>**Feature Importance Analysis**</u>

Another key benefit of selecting the Random Forest model is that it provides feature importance metrics. Understanding which features drive customer segmentation is crucial for the business as it enables better business insights and facilitates strategic decision-making.



- **Interpretability**

Although Random Forest is considered a black-box model, it is still more interpretable than some complex models, such as neural networks. The ability to extract the importance of features and analyse decision paths makes the random forest model more transparent and interpretable.

Having considered the accuracy, computational efficiency, interpretability and all the above factors, it can be safely concluded that the most appropriate model for this analytical solution is the Random Forest model.

Analytical Solution

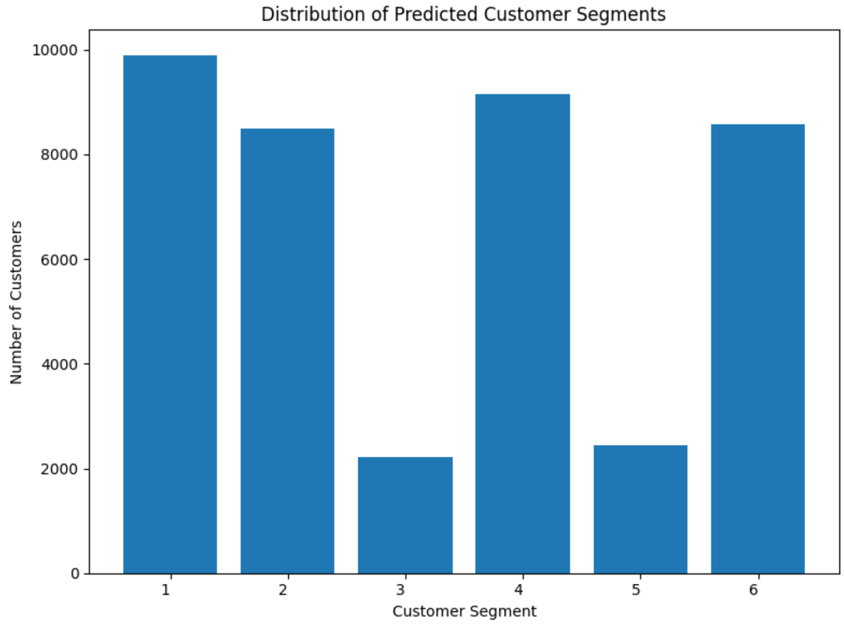
Using the final random forest model, the customer segment for the customers in the test csv has been implemented as shown below:

```
Final_Analytical_Solution = clean_test[['Customer_ID'
Final_Analytical_Solution
```

	Customer_ID	predicted_customer_segment
0	33574	1
1	10089	1
2	38329	1
3	11376	1
4	12410	1
...	...	...
40744	33698	5
40745	4185	5
40746	28664	5
40747	11874	5
40748	11431	5

40749 rows × 2 columns

The following is a visualisation of the distribution of customers across the segments from the test csv. As it can be seen, the customers have been correctly identified and segmented into six different categories with the largest number of customers coming from the first segment. Furthermore, segments three and five can be seen to have the lowest customers as well indicating which target markets require more focus.


Distribution of Predicted Customer Segments
Number of Customers
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
Customer Segment

Challenges and Limitations

1. Large Dataset - Dealing with a large number of customers made interpreting EDA graphs hard due to the high number of data points.

2. Computational Resource Limitation - Due to the limited computational resources available, it was very time-consuming and difficult to run certain techniques such as grid search and cross-validation. Models take very long to run.

3. Model Overfitting - Due to the data set being less complex, random forest models with a large number of decision trees tend to overfit. Therefore, hyperparameters had to be tuned down.

Business Impact and Recommendations

The classified clusters provide a clear distinction between customers based on their spending behaviour. This can help increase the effectiveness of companies' marketing strategies. By understanding the underlying characteristics of each segment, marketing efforts can be conducted

Segment 1 - Practical bulk buyer :

This segment consists of high dry sales, compared to moderate luxury and fresh sales. For this segment, we can expand on the varieties of goods in terms of dry sales, such as varieties of rice, varieties of canned goods, etc. By diversifying the dry sales segment, customers will appreciate the options available in their preferred sales segment.

Segment 2 - Fresh only buyer:

This segment focuses significantly on fresh sales, as opposed to dry and luxury sales. This indicates the majority of customers in this segment are health-conscious and prefer organic purchases. To attract these customers, we can highlight the quality of the product and the freshness of the goods. We can also provide promotions on this segment to encourage sales in this segment.

Segment 3 - Premium all round buyers :

In this segment, sales across all segments are fairly equal and high. This group of customers is a loyal customer base as they do the majority of their shopping in the supermarket. To encourage more purchases across the three segments, we can implement a loyalty reward system and maintain good customer service to ensure a good relationship is maintained with the customers.

Segment 4 - Necessity bulk Buyers :

This segment focuses heavily on dry sales as opposed to fresh and luxury sales. This indicates that the majority of customers can most likely be middle or lower-middle class. Therefore, we can focus on promotions for packages as demand for this customer group is price elastic. We can also use price-sensitive messages such as “save more with bulk deals”.

Segment 5 - all-around buyers :

In this segment, sales across all segments are equal but not as high as in segment 3. This can suggest that customers in the segment are not as frequent shoppers. In order to change that and make them regular customers we can increase the convenience of their shopping experience. This can be done by implementing a mobile app or online platform where customers can shop easily and save time. This can increase their tendency to shop more in our organization.

Segment 6 - Selective Fresh buyers :

This segment mainly focuses on fresh sales but also has a moderate interest in luxury sales and minimal interest in dry sales. This would suggest the customer group focuses on fresh quality goods, and they won’t mind premium prices, unlike segment 2fresh-onlyly buyer). To encourage more purchases and help customers experience the quality of fresh luxury goods, we can set out sample stands. This way customers can have an idea of quality firsthand. This can increase their willingness to increase their purchases.

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Appendix

- GitHub Repository: *https://github.com/Insight-Syndicate/DSPL_GCW*

- Google Drive:

  https://drive.google.com/drive/folders/1XkRUF4MCrgqzis3v2C7Bx9GQaK5XgFZI?usp=sharing